

Optimal Stopping and Reputation

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Abstract

An agent decides if and when to take an irreversible action while learning dynamically whether it will succeed. The agent is reputation-driven, seeking to signal his ability to learn. In equilibrium, the agent engages in *contrarian* behavior: he is more willing to act on limited information when the prior probability of success is lower. Reputation induces systematic distortions in decision timing: the agent acts prematurely when it is highly disadvantageous ex-ante, and with delay when it is advantageous. Distortions can also dynamically reverse, with the agent acting prematurely at the beginning of the research process and with excess caution at its end.

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1. Introduction

Decision makers often decide how long to acquire information before taking an irreversible action. In such Wald problems, this stopping decision is often driven by an exogenous tradeoff between accuracy and timeliness.

In practice, however, such decisions are frequently made by agents who face reputational concerns and therefore have incentives to signal their competence. For example, a politician deciding whether to implement a new policy may consider how this decision and its timing affect public perceptions of her decision-making ability, as this can influence her re-election prospects. Similarly, managers are often tasked with deciding whether to make irreversible investments in new projects and technologies, and may be concerned with how the speed and success of such decisions reflect their ability to identify profitable new ventures. Indeed, the influence of CEO career concerns, and more specifically reputational concerns, on corporate investment decisions is empirically documented ([Graham, Harvey, and Rajgopal \(2005\)](#), [Nadeem, Zaman, Suleman, and Atawnah \(2021\)](#)). In such settings, accuracy and timeliness are of *endogenous* value to the agent in signaling her competence.

In this paper, I study an agent who takes an irreversible action under uncertainty. This agent seeks a *reputation for learning*, strategically stopping to signal a high learning ability. I pose two questions. First: what is the optimal stopping behavior of a reputation-driven agent? I consider how this behavior differs qualitatively from that of a standard Wald problem where reputational motives are absent. Second, I ask how this motive distorts decision timing, and how these distortions depends on the nature of the decision problem.

To this end, I present a model of optimal stopping with reputational motives. An agent (e.g., manager) dynamically learns about an underlying state, namely whether some irreversible action will succeed or fail, and decides when to stop learning and act before some exogenous deadline. The agent may be either good, receiving an informative signal in every period, or bad, receiving no information. The agent faces a combination of two objectives: a *decision payoff* which is an exogenous payoff from accurate and timely decision making as in a standard Wald problem, and a *reputational payoff* which is the belief of an outside observer

(e.g., the market) that he is of high ability. This belief is formed ex-post with knowledge of both the agent's stopping behavior and whether the action was a success or failure. I seek to understand how this reputational payoff affects the agent's behavior.

In equilibrium, the reputation-driven agent engages in *contrarian behavior*: he is more willing to act, and to act quickly, when success is ex-ante less likely. Precisely, at any time the agent will act for a larger range of posterior beliefs when the prior probability of success is lower. This stands in contrast to a standard Wald problem, where behavior is dictated solely by the posterior belief. Contrarian behavior is driven by the endogenous impact of speed on reputation: speed is reputationally rewarded conditional on a success but penalized conditional on a failure, and crucially, the relative magnitudes of this reward and penalty depend on the prior probability of success. When success is ex-ante unlikely, acting quickly and succeeding strongly signals the agent's learning ability, thus yielding a large reputational benefit. However, acting quickly and failing is relatively uninformative, implying a small reputational cost. This incentivizes early action. In contrast, when success is likely a priori, acting quickly on a success does little to signal the agent's ability, but acting quickly on a failure is highly reputationally damaging. The agent thus waits until he is highly confident that success will occur before acting in order to avoid this reputational harm, leading to delayed action.

I then more generally consider the distortionary effects that reputational motives have on the timing of action. I find that reputation can cause the agent to either act prematurely or with excess delay, and which type of distortion arises depends on the nature of the decision problem. Furthermore, distortions can dynamically change over the course of the agent's research process.

When the ex-ante value of acting is very low, reputation causes the agent to act prematurely at every stage of the learning process. This premature action is driven not just by the prior of success, but also on the payoffs from success and failure: the agent acts prematurely whenever the prior probability of success is low (reflecting the agent's contrarian motive), but also when the upside of success is small or the downside of failure is large. Intuitively, this is due to the fact that in such decision problems, acting is always highly sub-optimal for the bad agent's decision payoff. However, a good agent may potentially learn over time

that success will indeed occur, making acting optimal. In equilibrium, acting, and acting quickly, thus serves as a signal that the agent is good at learning, one that is costly for the bad agent profit-wise and thus credible. This induces premature action in every period. Conversely, when the ex-ante value of acting is high, the opposite effect arises: reputation induces delay, or excess caution. Under a high ex-ante value of acting, acting immediately is always the optimal course of action for the bad agent's decision payoff, while the good agents may acquire information leading them to believe that failure is likely, making abstention optimal. So in this case, abstention serves as a costly signal that the agent is good, leading to delayed action.

In the context of irreversible investment, these results suggest that reputational motives should induce over-investment in projects that are long shots and under-investment in ones that are safer or more promising. Such a relationship between the probability of success and inefficiency in investment has been empirically studied. [Pammolli, Magazzini, and Riccaboni \(2011\)](#) document a decline in R&D productivity among pharmaceutical firms in the US and Europe from 1990 to 2008. They provide evidence that this decline in productivity is driven not by a fall in productivity within classes of drugs, but rather a reallocation of R&D expenditures away from drugs with a high probability of success to those with a low probability of success. These distortionary results suggest that such investment behavior may partly be explained by reputational motives.

I then consider how distortions can be dynamic in nature. When the probability of success takes an intermediate value, the agent can switch from premature action early in the research process to a reluctance to act later on. Such reversals happen when discounting is slow (i.e., the agent is patient). In such scenarios, the optimal rule in the absence of reputation entails acting in early periods only if very confident that success will occur, as the option value of waiting is high. However, since speed is reputationally rewarded the agent responds by acting prematurely in early periods. As there is increasingly less time left for research, the option value of waiting declines. Such late action indicates that the agent's early signals were not highly informative and thus can be reputationally damaging, making the agent overly reluctant to act. This suggests that reputational motives can make managers susceptible to inadequately responding to the information they can

gather, by either investing before they have had allowed sufficient time for research or refraining from investing once their research has revealed that investment is likely profitable.

This paper contributes to the literature on real options models of investment, in which a profit-maximizing decision maker chooses when to make an irreversible investment under uncertainty. In canonical settings ([Dixit and Pindyck \(1994\)](#), [McDonald and Siegel \(1986\)](#)), this uncertainty concerns future flow payoffs, but not the underlying data generating process. In contrast, I consider a decision maker who is uncertain about the profitability of investment and whose option value arises from the ability to learn. A subset of this literature incorporates dynamic learning ([Bernanke \(1983\)](#), [Cukierman \(1980\)](#), [Décamps, Mariotti, and Villeneuve \(2005\)](#)). In [Bernanke \(1983\)](#) and [Cukierman \(1980\)](#), a decision maker chooses one of several projects to pursue while learning about their relative profitabilities. Meanwhile, [Décamps et al. \(2005\)](#) consider a single project whose flow payoffs follow a Brownian motion with unknown drift, where the decision maker learns about this drift by observing past returns. They find that the optimal stopping rule can entail investing after a drop in expected returns. Rather than studying the profit-maximizing behavior, I consider how a desire to signal learning ability can lead to deviations from such optimal behavior.

Indeed, there is an expansive literature studying the effect of agency issues on timing decisions. In [Grenadier and Wang \(2005\)](#), an agent makes an irreversible investment under both moral hazard and adverse selection, while in [Grenadier, Malenko, and Malenko \(2016\)](#) an agent communicates private information about investment profitability while having a bias towards early or late investment. A subset of this literature specifically considers an agent who uses their timing decision for signaling purposes. In [Bobtcheff and Levy \(2017\)](#) and [Bouvard \(2014\)](#), an entrepreneur decides how long to experiment before investing in a project, where investment timing signals project quality and thus affects their ability to obtain outside funding. [Bouvard \(2014\)](#) finds that investment is delayed under the optimal contract compared to first best, while [Bobtcheff and Levy \(2017\)](#) find that in a contract-free environment, investment is hasty when learning is fast and delayed otherwise. Meanwhile, [Grenadier and Malenko \(2011\)](#), [Thomas \(2019\)](#), and [Halac and Kremer \(2020\)](#) study an agent who derives utility directly from

market perceptions of project quality. [Grenadier and Malenko \(2011\)](#) provide a general model of such signaling in a real options setting. In [Thomas \(2019\)](#) and [Halac and Kremer \(2020\)](#), an agent learns dynamically about the quality of a project and signals quality via a decision to abandon it. Both papers document inefficiently delayed abandonment. [Halac and Kremer \(2020\)](#) further find that the magnitude of this distortion is increasing in the prior belief that the project is profitable. I contribute to this literature by considering an agent who seeks to signal learning ability, i.e., the capacity to identify project quality, rather than project quality itself. This difference in objective gives rise to qualitatively different distortions. For one, reputation for learning induces contrarian behavior, wherein investment is premature for projects that are unlikely to succeed ex-ante and delayed for those that likely to succeed. Furthermore, this desire to signal learning ability can lead to dynamic reversals in distortions, a phenomenon which to my knowledge does not arise elsewhere in this literature.

Finally, this paper connects to the literature on reputation for learning, in which agents communicate or act to influence beliefs about their learning ability. First studied by [Scharfstein and Stein \(1990\)](#), [Ottaviani and Sørensen \(2006a\)](#) present a general static model of cheap talk under reputation for learning, finding that truthful revelation of information is generically impossible. In a separate paper ([Ottaviani and Sørensen, 2006b](#)), the authors show that the prior affects the nature of such distortions: the sender shadows their messages towards the prior mean. More similar to the setting I consider, [Prendergast and Stole \(1996\)](#) and [Dasgupta and Prat \(2008\)](#) present dynamic models of investment and trading, respectively, where agents are motivated by both profit and reputation for learning. In [Prendergast and Stole \(1996\)](#), the agent dynamically adjusts their level of investment as they learn what the optimal investment position is. Reputation for learning induces the agent to overreact to new information at the beginning of this learning process, and under-react to it later on. I contribute to this literature by taking a real-options approach, i.e., considering the effect of reputation for learning on the timing of action. Methodologically, whereas these papers assume the agent is myopic, the simpler strategy space this timing decision induces allows one the tractability to consider a forward-looking agent, a fact that is central to the characterization. Like [Prendergast and Stole \(1996\)](#), I find that reputation can induce reversals in the nature of distortions. However, in the real-options setting I

study, this reversal pertains to the agents' willingness to commit to an irreversible decision. More recently, [Vong \(2025\)](#) considers a static setting, characterizing conditions under which the sender can influence the decisions of the receiver. This paper assumes that the capable sender observes a non-parametric, continuous signal as I do, and further allows the incapable sender to observe informative signals. [Deb, Pai, and Said \(2018\)](#) characterize the optimal contract to screen agents for ability, when agents dynamically learn and report about an unknown state.

The rest of the paper proceeds as follows. Section 2 presents the model. In sections 3 and 4, I characterize equilibrium strategies and reputation, respectively, establishing the agent's contrarian motive. In section 5, I characterize equilibrium timing distortions. In section 6, I present extensions of the model. Finally, section 7 concludes. All proofs are relegated to the appendix.

2. Model

There is one agent and one principal. Time $t \in \{1, \dots, T\}$ is discrete, with a finite horizon $T < \infty$.¹ The state $\theta \in \{0, 1\}$ denotes whether some action will be a success ($\theta = 1$) or failure ($\theta = 0$). The agent and principal have a common prior $p_0 = \Pr(\theta = 1) \in (0, 1)$. The agent is of type $i \in \{G, B\}$ (*good* or *bad*), which is time-invariant and independent of θ . The agent knows his type, but the principal does not and holds a prior $R_0 \equiv \Pr(i = G) \in (0, 1)$.

Learning and Acting The type determines the agent's ability to learn about θ . At the beginning of each t , a good agent observes some signal $s_t \in (0, \infty)$, distributed according to conditional density $f(\cdot|\theta)$. The signals s_t are labeled as their likelihood ratios, i.e., $s_t = \frac{f(s_t|\theta=1)}{f(s_t|\theta=0)}$. The s_t are i.i.d. across t given θ and $f(\cdot|\theta)$ is full support on $(0, \infty)$. Meanwhile, a bad agent has no ability to learn: he observes no signals.²

The agent chooses if and when to act. Specifically, at each t (after observing s_t if $i = G$), the agent chooses $a_t \in \{\emptyset, 1\}$. $a_t = 1$ denotes *act* while $a_t = \emptyset$ denotes *abstain*. Acting is irreversible: if $a_t = 1$, then $a_s = 1$ for all $s > t$. Otherwise the

¹ In [Section 6](#), I extend the model to an infinite horizon setting ($T = \infty$).

² The assumption that the bad agent entirely is unable to learn is made for simplicity. Indeed, in [Section 6](#), I show that the main results are robust to allowing the bad agent limited learning ability.

agent chooses freely. Let $\tau \in \{1, 2, \dots, T, \emptyset\}$ denote the time at which the agent acts (i.e., the first t where $a_t = 1$), with $\tau = \emptyset$ denoting that the agent never acts.

Reputation and Payoffs The principal observes the time of action τ and the state θ and forms a posterior belief about the agent's type, which is the agent's *reputation*. This belief is given by a reputation function $R : \{1, \dots, T, \emptyset\} \times \{0, 1\} \rightarrow [0, 1]$, where $R(\tau, \theta)$ is the principal's belief that $i = G$ after observing τ and θ .³

The agent is motivated by two objectives: (1) accurate and timely decision-making and (2) reputation. Formally, the agent's payoff is given by:

$$U(\tau, \theta) = (1 - X)D(\tau, \theta) + XR(\tau, \theta),$$

where $D(\tau, \theta)$ is the *decision payoff* and $X \in [0, 1]$. The decision payoff is given by

$$D(\tau, \theta) = \beta^\tau K_\theta \mathbb{I}(\tau \neq \emptyset),$$

where $K_1 > 0 > K_0$ and $\beta \in (0, 1)$.

Under this payoff function, the agent's payoff is a combination of two components. The first component, the decision payoff, captures the exogenous benefit from accurate and timely decision-making, as in a standard Wald problem. Namely, acting is beneficial if and only if $\theta = 1$ and the payoff from acting is geometrically discounted at rate β . In the context of irreversible investment, this decision payoff can be interpreted as the profit from investment. The second component is the agent's reputation, and the parameter X specifies the relative weight the agent places on reputation.

2.1. Equilibrium

A strategy for the good agent is a sequence of functions $\{A_t^G\}_{t=1}^T$, where $A_t^G : [0, 1] \rightarrow [0, 1]$ and $A_t^G(p)$ denotes the probability of acting at time t (choosing $a_t = 1$) given that the agent has not yet acted (i.e., given $a_s = \emptyset$ for all $s < t$) and holds belief $p = \Pr(\theta = 1)$.⁴ A strategy for the bad agent $A_t^B \in [0, 1]$ denotes

³One could alternatively assume that the state θ is observed by the principal only if the agent acts, and the below analysis would remain qualitatively unchanged.

⁴In general, strategies could depend on the entire sequence of signals the good agent receives. However, it is without loss to restrict attention to such Markov strategies within the class of

the probability of acting at t under belief p_0 . For any signal history $(s_1, \dots, s_t)$, $P(s_1, \dots, s_t)$ is the good agent's posterior belief that $\theta = 1$.

I seek a Markov perfect equilibrium. This consists of strategies $\{A_t^i\}_{t=1}^T$, a reputation function R and beliefs P such that the strategies maximize $E_\theta[R(\tau, \theta)]$ at all (t, p) and P, R are consistent with Bayes rule given the strategies.

Selection Because reputation is an equilibrium object, there exist equilibria one may deem unintuitive whenever the reputational motive (X) is sufficiently large. These can entail babbling or other preverse behavior in which the good agent signals his ability by acting on failures rather than successes.

I thus impose two selection criteria. First, the equilibrium must be informative about the agent's type in every period, when possible. Precisely, the good and bad agent cannot have the same joint distribution of action and state conditional on not yet having acted, which allows the principal to make inferences about the agent's type. This rules out babbling equilibria, and is stated as the informativeness assumption ([Assumption 1](#)).

Assumption 1 (Informativeness). *At any t , if $A_s^B \neq 1$ for all $s < t$, then for some θ :*

$$Pr(\tau = t | i = G, \theta, \tau \not\leq t) \neq Pr(\tau = t | i = B, \theta, \tau \not\leq t).$$

Next, I impose that the agent cannot be rewarded for minimizing the decision payoff. This rules out “reverse signaling” equilibria where the agent only acts he is sufficiently sure that $\theta = 0$, because doing so signals learning ability. This is formalized as [Assumption 2](#).

Assumption 2 (No reverse signaling). *In any t the following does not hold for any $\tau \in \{t + 1, \dots, T, \emptyset\}$:*

$$R(t, \theta = 0) > R(\tau, \theta = 0) \text{ and } R(t, \theta = 1) < R(\tau, \theta = 1).$$

One may ask how these criteria relate to more standard criteria in signaling equilibria that satisfy the selection criteria specified below.

games. While a generalization of $D1$ ⁵ could be used instead of informativeness, it would not rule out reverse signaling equilibria, as such equilibria exist even in the absence of off-path behavior. Rather, one can interpret these two criteria as ruling out behavior that is not robust to a moderate concern for the decision payoff.⁶

3. Equilibrium behavior and prior beliefs

I now characterize the agent's decision-making behavior in equilibrium. I begin by establishing equilibrium existence and showing that strategies take a simple form. I then present and compare two benchmark cases to clearly illustrate the impact of reputational motives on decision making behavior: one where agent places no weight on reputation ($X = 0$) and the other where the agent is purely reputation-driven ($X = 1$). In this comparison, I begin by considering the static environment ($T = 1$) and later the general, dynamic one.

3.1. Equilibrium structure

Here, I show that in any equilibrium, the good agent plays cutoff strategies in every period. To this end, I begin by establishing two useful properties of the agent's equilibrium value function.

Let $V_t^i(p, a)$ denote the type- i agent's time- t value from playing $a_t = a \in \{\emptyset, 1\}$, given he has not yet acted (i.e., $a_s = \emptyset$ for all $s < t$). I first establish that at any time, the value of abstaining (i.e., the continuation value) must be convex in the belief about the state. This follows from Blackwell's theorem, and is stated as [Lemma 1](#).

Lemma 1. $V_t^G(p, \emptyset)$ is convex in p for all $t \in \{1, \dots, T\}$.

Next, I establish that an agent who knows that $\theta = 1$ strictly prefers acting, while an agent who knows that $\theta = 0$ strictly prefers abstaining. That is, at extreme beliefs, the agent behaves in a way that is consistent with an optimal decision maker (one who faces $X = 0$). This property is stated as [Lemma 2](#).

⁵ $D1$ must be appropriately generalized to be applied here, as the agent's payoff is directly a function of the principle's belief, rather than her action.

⁶ Indeed, one can show that if X lies below some threshold, both assumptions must be satisfied in any equilibrium.

Lemma 2. *In any equilibrium, in all t :*

$$V_t^G(1, 1) > V_t^G(1, \emptyset) \text{ and } V_t^G(0, \emptyset) > V_t^G(0, 1). \quad (1)$$

This follows from the above two selection assumptions, and can most easily be illustrated under a pure reputational motive ($X = 1$). Informativeness implies that the value of acting and abstaining depend on the true state. Meanwhile, the no reverse signalling assumption ensures the agent's value cannot be maximized via behavior which is exactly opposed to optimal decision making. Thus, the dependence of the value on the state is consistent with optimal decision-making.

I now establish equilibrium existence, and show that the good agent must play cutoff strategies in equilibrium. This is stated as [Proposition 1](#).

Proposition 1. *There exists an equilibrium. In any equilibrium, the good agent plays cutoff strategies in every period: for all t , there exists $p_t^* \in (0, 1)$ such that*

$$A_t^G(p) = \begin{cases} 0 & \text{for all } p < p_t^* \\ 1 & \text{for all } p > p_t^*. \end{cases}$$

Existence follows by applying the Kakutani Fixed Point Theorem to a constrained space of strategy profiles employed by the two types of agent. [Proposition 1](#) further establishes that at all times, there exists an interior, time-dependent cutoff belief such that the good agent acts if and only if his belief lies above this cutoff.

This result can be illustrated by a geometric argument. Figure 1 plots, for any given t , $V_t^G(p, 1)$ and $V_t^G(p, \emptyset)$, i.e., the good agent's value from acting and abstaining, respectively. Now, let us make two observations. First, $V_t^G(p, 1) = pR(t, 1) + (1 - p)R(t, 0)$ is linear in the belief p , while $V_t^G(p, \emptyset)$ is convex in p ([Lemma 1](#)). Second, [Lemma 2](#) implies that $V_t^G(p, 1)$ lies strictly above $V_t^G(p, \emptyset)$ when $p = 1$ and strictly below $V_t^G(p, \emptyset)$ when $p = 0$. Together, these two facts imply that $V_t^G(p, 1)$ intersects $V_t^G(p, \emptyset)$ at a unique interior point p_t^* , and thus $V_t^G(p, 1) > (<) V_t^G(p, \emptyset)$ to the right (left) of this point.

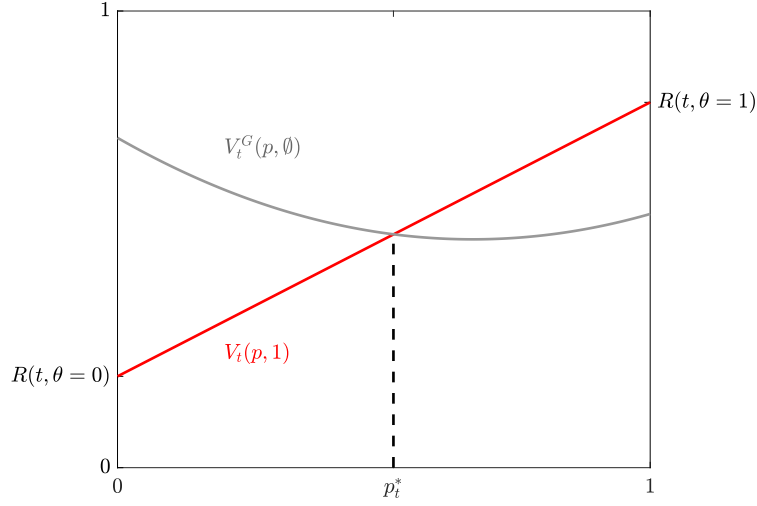


Figure 1: The good agent's value of acting ($V_t(p, 1)$) and abstaining ($V_t(p, \emptyset)$), as a function of his belief p .

3.2. Benchmark: Decision-optimal rule

As a benchmark, I characterize the optimal stopping rule without any reputational motive (i.e., when $X = 0$). I refer to this as the *decision-optimal rule*. This is stated as [Proposition 2](#).

Proposition 2. *The decision-optimal rule is the following:*

- *The good agent plays a cutoff rule in every period:*

$$A_t^G(p) = \begin{cases} 1 & \text{if } p > \hat{p}_t \\ 0 & \text{if } p < \hat{p}_t, \end{cases} \quad (2)$$

where the $\hat{p}_t \in (0, 1)$ are unique, strictly decreasing in t , and independent of p_0 .

- *The bad agent acts in any period if $p_0 > \hat{p} \equiv \frac{-K_0}{K_1 - K_0}$ for all t :*

$$A_t^B = \begin{cases} 1 & \text{if } p > \hat{p} \\ 0 & \text{if } p < \hat{p}. \end{cases}$$

[Proposition 2](#) states that the good agent employs a cutoff rule in every period, where the cutoffs strictly decrease over time. The decreasing nature of these cutoffs

is due to the fact that, as the deadline T approaches, the agent has less time left to learn, and thus has a lower option value at any given belief. Hence, it will be optimal to act for a wider range of beliefs as time passes. Notably, while these cutoffs will depend on the agent's learning process and payoff parameters (K_0 and K_1), they are independent of the prior belief p_0 . This is due to the fact that given any exogenous payoff function (such as D), the prior impacts the agent's optimal decision only via the posterior belief at any time. That is, the prior is intrinsically irrelevant to the agent's decision-making. As I will soon illustrate, this is not true for a reputation-maximizing agent.

Meanwhile, the bad agent acts immediately if acting has a positive net present value under the prior, and otherwise never acts. This is because there is discounting and the bad type is unable to learn: if his prior is such that acting is optimal, he will do so immediately and otherwise abstains indefinitely.

3.3. Static characterization under pure reputational motive

I now characterize the equilibrium when the agent is purely reputation-driven ($X = 1$). I begin with the static environment ($T = 1$), where the qualitative departures from the decision-optimal benchmark are most simply illustrated⁷ and then consider the dynamic environment. Throughout this subsection, I omit the time index from all objects.

Claim 1 characterizes the equilibrium in the case where $X = 1$ and $T = 1$. It states that the bad agent must mix between acting and abstaining, while the good agent acts if and only if his posterior exceeds the prior.

Claim 1 (Static contrarianism). *When $X = 1$ and $T = 1$, there exists a unique equilibrium. Under this equilibrium, the bad agent mixes ($A^B \in (0, 1)$) and the good agent plays the cutoff strategy $p^* = p_0$.*

Let us begin by considering why the bad agent must mix. **Proposition 1** established that the good agent plays an interior cutoff strategy $p^* \in (0, 1)$. Since the good agent's signals are full support over the likelihood ratios, it follows that the good agent both acts and abstains with positive probability. If the bad agent were to, for

⁷This static case is also a special case of [Vong \(2025\)](#)'s static model, which characterizes the informativeness of communication rather than the strategies of the players, as I do here.

instance, only act in equilibrium ($A^B = 1$) then only the good type would abstain with positive probability, and thus the reputation function would assign a perfect reputation to an agent who abstains regardless of the realization of the state. A purely reputation-driven agent could then profitably deviate by always abstaining. By analogous reasoning, if the bad agent were to only abstain in equilibrium ($A^B = 0$), acting would be a profitable deviation. We conclude that the bad agent must mix in equilibrium.

Meanwhile, the good agent's strategy follows from the fact that, in a static setting, the good and bad agent have identical (continuation) values: $V_t^G(p, a) = V_t^B(p, a)$ for all beliefs p and $a \in \{\emptyset, 1\}$. Because there is a unique belief at which the good agent is indifferent between acting and abstaining, and the bad agent must be indifferent at p_0 to sustain mixing, the good agent's cutoff must equal the prior.

This result suggests a fundamental, qualitative difference between the decision-making behavior of a reputation maximizer compared to an agent maximizing an exogenous payoff function: while the prior belief does not impact the behavior of an agent maximizing decision payoffs, the prior itself is crucial to the reputation-driven agent's behavior. Namely, the reputation-driven agent's behavior is *contrarian*: he is more willing to act (formally, will act for a larger range of beliefs) when success is ex-ante less likely. This holds despite the fact that the prior itself provides no payoff-relevant information beyond what captured by the posterior p . Rather, the prior impacts the agent's equilibrium behavior via the reputation function: lowering the prior probability of success leads the equilibrium reputation function to further reward acting (rather than abstaining), thus making the agent more willing to act. The relationship between the prior and equilibrium reputation function will be more precisely explored in [Section 4](#).

3.4. Dynamic characterization under pure reputational motive

Having characterized the static equilibrium, I now consider a dynamic environment. In the dynamic equilibrium, the prior belief serves as a lower bound on the cutoffs the good agent employs in every period. This result is stated as [Proposition 3](#).

Proposition 3. *Suppose $X = 1$. In any equilibrium,*

- $A_t^B \in (0, 1)$ for all t .

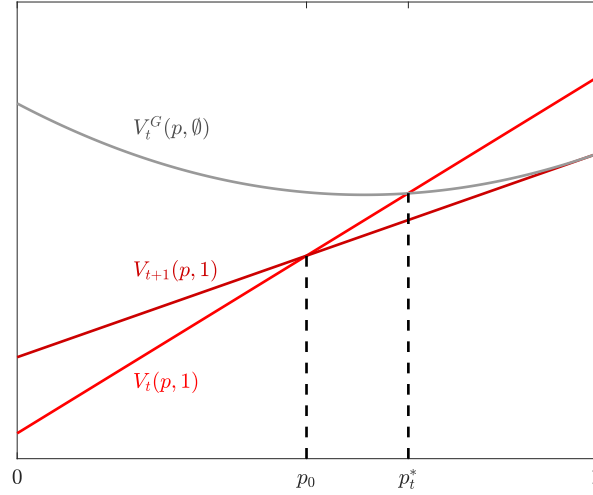


Figure 2: The good agent's point of indifference, p_t^* , lies strictly to right of the prior, p_0 .

- $p_T^* = p_0$ and $p_t^* > p_0$ for all $t < T$.

The proposition first establishes that the bad agent mixes in every period between acting and abstaining, which follows from the same reasoning as in the static case. Meanwhile the good agent employs a dynamic cutoff strategy. In the final period T , this cutoff equals the prior p_0 , by the same reasoning as in the static case. However, for periods prior to the deadline ($t < T$), the cutoff p_t^* strictly exceeds the prior. This is due to the fact that the good agent observes informative signals about the state in future periods and thus enjoys a strictly higher continuation value than the bad agent.

This result can more precisely be illustrated by a geometric argument. Figure 2 plots the good agent's value as a function of his beliefs, as in Figure 1, for some $t < T$. Figure 2 also plots the value from acting in the next period, $V_{t+1}(p, 1)$. This value lies strictly below the good agent's continuation value at time t , $V_t^G(p, \emptyset)$ for all $p < 1$. This is because a good agent who continues in $t + 1$ can at least obtain the value from acting in $t + 1$, and a strictly higher value by updating his belief using his new signal s_{t+1} and only acting when it is optimal to do so. Meanwhile, $V_t(p, 1)$ and $V_{t+1}(p, 1)$ must intersect at p_0 : this is because the bad agent mixes, which means that he must be indifferent between acting in t and $t + 1$. It follows that the good agent's point of indifference p_t^* , which is where $V_t^G(p, \emptyset)$ and $V_t(p, 1)$ intersect, lies strictly to the right of p_0 .

This result implies that a higher prior will yield a strictly higher cutoff in some periods (e.g., T), but not necessarily all. However, one can show that a sufficiently large increase in the prior belief yields strictly higher cutoffs in all periods. This is stated formally as [Proposition 4](#).

Proposition 4 (Dynamic contrarianism). *Assume $X = 1$. Fixing all other parameters, for any p_0 there exists $\bar{p} \in (0, 1)$ such that if $p'_0 < \bar{p}$,*

$$p_t^{*'} < p_t^*$$

for all t and any equilibrium strategies $(p_t^)_{t=1}^T$ and $(p_t^{*'})_{t=1}^T$ under p_0 and p'_0 , respectively.*

[Proposition 4](#) shows that contrarianism has dynamic implications: the agent not only acts more often, but more hastily when success is ex-ante less likely. Put differently, a lower probability of success yields the agent less cautious, more willing to prioritize early action over further experimentation. In the following section, I consider which equilibrium forces induce the agent to behave in this way.

4. Reputation: speed and accuracy

In this section, I characterize the equilibrium reputation function. This elucidates the equilibrium incentives at play and why the agent engages in contrarian behavior. To most clearly illustrate these incentives, I again restrict attention to a purely reputation-driven agent ($X = 1$).

I begin by characterizing the impact of speed on the agent's reputation. Formally, *speed* refers to acting in an earlier period, holding fixed the state. One can show that speed improves an agent's reputation conditional on success ($\theta = 1$) but is reputationally damaging if failure occurs ($\theta = 0$). That is, early errors (acting when $\theta = 0$) are worse for reputation than late ones. This is stated as [Proposition 5](#).

Proposition 5 (Speed). *In any equilibrium when $X = 1$, for all t :*

- $R(t, \theta = 1)$ is strictly decreasing in t ,
- $R(t, \theta = 0)$ is strictly increasing in t .

The reason for this conditional effect of speed lies in the fact that the bad agent mixes in equilibrium: if speed is reputationally beneficial conditional on success, it must be costly conditional on failure. Otherwise, the bad agent could profitably deviate by always acting sooner.

While [Proposition 5](#) establishes that speed has a positive effect on reputation when $\theta = 1$ and a negative effect when $\theta = 0$, it does not speak to their magnitudes. In fact, the relative magnitudes of these effects are dictated by the prior p_0 : the higher the prior, the lower the positive effect of speed when $\theta = 1$ compared to the negative effect when $\theta = 0$. I will argue that this relationship explains why the agent engages in contrarianism in equilibrium. The result is formalized as [Corollary 1](#).

Corollary 1 (Speed and the prior). *In equilibrium when $X = 1$, for all $t < T$:*

$$\frac{R(t, 1) - R(t + 1, 1)}{R(t + 1, 0) - R(t, 0)} = \frac{1 - p_0}{p_0}.$$

There is a concise explanation for this result: if the prior increases, $\theta = 1$ is more likely to realize ex-ante. To preserve the bad agent's indifference between acting in t and $t + 1$, the benefit of speed when $\theta = 1$, $R(t, 1) - R(t + 1, 1)$, must decrease compared to the cost of speed when $\theta = 0$, $R(t + 1, 0) - R(t, 0)$, to compensate for the fact that speed is more likely to be beneficial. It can also be illustrated graphically: [Figure 3](#) plots the agent's value from acting in period t and $t + 1$, under a low prior (left panel) and high prior (right panel). These lines have endpoints that correspond to realizations of reputation and must intersect at the prior. When the prior is low, speed is highly beneficial for reputation when acting correctly (under $\theta = 1$) but minimally harmful when acting erroneously (under $\theta = 0$). Conversely, when the prior is high, speed is minimally beneficial when acting correctly, but highly harmful when acting erroneously.

Such a relationship incentivizes contrarian behavior. When the prior is high, speed provides only a minor reputational benefit if success occurs, but can inflict substantial reputational harm in the event of failure. Thus, the agent is incentivized to gather more information to be sufficiently sure that success will occur before acting, leading to delay. Meanwhile, when the prior is low, speed provides a substantial reputational benefit if success occurs, but minimal harm if

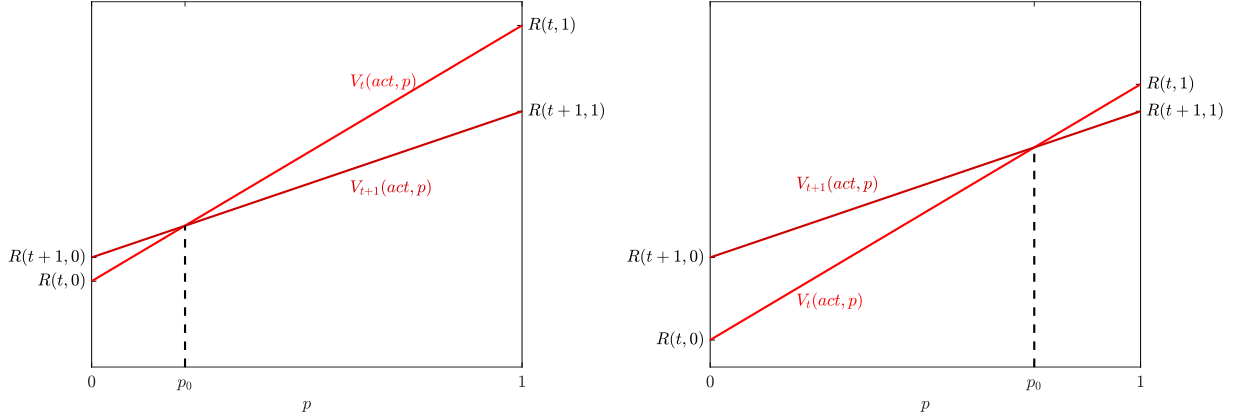


Figure 3: The value from acting in t and $t + 1$ under a low prior (left panel) and high prior (right panel).

failure occurs. This incentivizes the agent to act quickly even if lacking confidence that success will occur. Together, these incentives induce contrarian behavior.

While we have thus far considered speed, one may ask what effect accuracy has on reputation. Formally, *accuracy* refers to acting on a success ($\theta = 1$) rather than a failure ($\theta = 0$), fixing the time of action. While accuracy always benefits the agent, it is more beneficial in earlier periods.

Corollary 2. *In equilibrium, $R(t, 1) - R(t, 0) > 0$ and is strictly decreasing in t .*

This is an immediate corollary of [Proposition 5](#), and can be seen from either panel of [Figure 3](#): acting is more reputationally beneficial conditional on success and reputationally damaging conditional on failure in early periods. Thus, while accuracy is always beneficial to the agent, it is most consequential in early periods.

5. Timing distortions

I now consider how reputation distorts the timing of action compared to the decision-optimal benchmark. I find that reputation can induce both premature and delayed action depending on the nature of the decision problem. The agent acts prematurely when acting is particularly disadvantageous ex-ante which can be driven by a low probability of success, reflecting the agent's contrarian motive, but also by a low benefit of success (K_1) or high cost from failure (K_0). Conversely,

the agent acts with delay when acting is favorable ex-ante. When discounting is sufficiently slow, reputation can induce dynamic reversals in distortions wherein the agent acts prematurely at the beginning of his learning process and with excess caution towards its end.

5.1. Premature action

I begin by characterizing the conditions under which the agent acts prematurely throughout his learning process. Precisely, *premature action* occurs at t if $p_t^* < \hat{p}_t$, i.e., the good agent acts under a strictly larger range of beliefs in equilibrium than the decision-optimal rule. Throughout this characterization, I relax the assumption that the agent is purely reputation-driven, and instead consider an arbitrary X . It is thus helpful to decompose the equilibrium value function into two components:

$$V_t^i(p, a) = (1 - X)V_t^{D,i}(p, a) + XV_t^{R,i}(p, a),$$

where $V_t^{D,i}$ denotes the agent's *decision value* and $V_t^{R,i}$ denotes his *reputational value*:

$$\begin{aligned} V_t^{D,i}(p, a) &\equiv E_{\tau, \theta}[D(\tau, \theta) | (A_s^i)_{s=t+1}^T, a_t = a, i] \\ V_t^{R,i}(p, a) &\equiv E_{\tau, \theta}[R(\tau, \theta) | (A_s^i)_{s=t+1}^T, a_t = a, i], \end{aligned}$$

and $(A_s^i)_{s=1}^T$ denotes the agent's equilibrium strategy. Notably, both $V_t^{D,i}$ and $V_t^{R,i}$ condition on the agent playing their equilibrium strategy in the continuation game. Because the agent chooses their equilibrium strategy to maximize a convex combination of his decision value and reputational value, this strategy does not necessarily maximize either his decision payoff and reputational payoff alone. This implies that while the equilibrium decision value and decision-optimal value of acting are equivalent, the decision value of abstaining may in general be less than that of the decision-optimal value. I.e., letting $\hat{V}$ denote value function under $X = 0$, in general:

$$V_t^{D,G}(p, 1) = \hat{V}_t^G(p, 1) \text{ and } V_t^{D,G}(p, \emptyset) \leq \hat{V}_t^G(p, \emptyset). \quad (3)$$

Before proceeding, let the *net decision (reputational) value of acting* $W_t^{D,i}(p)$

$(W_t^{R,i}(p))$ denote the difference in value from acting and abstaining:

$$W_t^{D,i}(p) \equiv V_t^{D,i}(p, 1) - V_t^{D,i}(p, \emptyset) \quad W_t^{R,i}(p) \equiv V_t^{R,i}(p, 1) - V_t^{R,i}(p, \emptyset).$$

I first show that premature action occurs whenever acting has positive net reputational value at the decision-optimal cutoff. This is stated as [Lemma 3](#).

Lemma 3. *Suppose $X > 0$. In any equilibrium, for any t : if $W_t^{R,G}(\hat{p}_t) > 0$, then the good agent acts prematurely in t .*

At the decision-optimal cutoff $\hat{p}_t$, the expected value from acting equals the decision-optimal value of abstaining: $\hat{V}_t^G(p, 1) = \hat{V}_t^G(p, \emptyset)$. Thus by (3), the net decision value of acting must be weakly positive in equilibrium: $W_t^{D,G}(\hat{p}_t) \geq 0$. If the agent has a strictly positive net reputational value at $\hat{p}_t$, then for any positive weight on reputation, there must be a strictly positive net *total value* of acting at $\hat{p}_t$. Thus, the equilibrium point of indifference lies to the left of $\hat{p}_t$: $p_t^* < \hat{p}_t$.

I now establish conditions on the decision problem in which reputation leads to premature action.

Theorem 1 (Premature action). *Suppose $X > 0$. There exist thresholds $\bar{p} \in (0, 1)$, $\underline{K}_0$, $\bar{K}_1$ such that the agent acts prematurely in all t when any one of the following holds:*

$$p_0 < \bar{p} \quad K_0 < \underline{K}_0 \quad K_1 < \bar{K}_1,$$

holding fixed all other parameters.

[Theorem 1](#) states that the agent acts prematurely whenever investment is sufficiently suboptimal ex-ante. Such sub-optimality could be driven by a low prior ($p_0 < \bar{p}$), meaning that the agent distorts the timing of their action in accordance with the contrarian motive established above. However, other factors can also drive such suboptimality: fixing a prior, the agent will also act prematurely whenever the payoff parameters align with acting being sub-optimal. This can entail a low benefit from acting on a success ($K_1 < \bar{K}_1$) or a large cost from acting on a failure ($K_0 < \underline{K}_0$).

The intuition behind this result lies in the bad agent's incentives. In equilibrium, any behavior that is reputationally rewarded—formally, any (τ, θ) pair yielding a reputation exceeding the prior reputation R_0 —must be exhibited less often by

the bad agent than the good agent. However, the bad agent faces two objectives: optimal decision-making and reputation maximization.⁸ So, behavior that is reputationally rewarded must be costly *in expectation* for his decision payoff; otherwise he would always engage in such behavior, and it would not serve as signal that the agent is good. When acting is highly suboptimal ex-ante (due to a low p_0 , low K_1 or low K_0), the reputation function thus rewards acting in every period regardless of the realized state θ . This induces the good agent to act at beliefs below the decision-optimal cutoffs, leading to premature action.

This result holds not only when the agent is highly reputation-driven, but for any weight $X > 0$ on reputation. Thus, it can be explained more precisely by considering two different cases: one where reputational motives are small (low X) and the other where they are large (high X). When the reputational motive is small and acting is sufficiently suboptimal ex-ante for the decision payoff, the bad agent will never act in equilibrium. This is because the negative decision value from acting at t , $W_t^{D,B}(p_0)$, outweighs any possible gain in reputational value, $W_t^{R,B}(p_0)$. The good agent, however, may observe signals strongly indicating $\theta = 1$, making acting optimal with positive probability. Since only the good agent acts, the reputation function must assign a perfect reputation to acting regardless of θ . By [Lemma 3](#), this implies premature action.

When the reputational motive is large, the bad agent does not exclusively abstain in equilibrium but instead mixes (as was true for the purely reputation-driven agent in Section 3). Thus, the bad agent is indifferent between acting and abstaining at any t even though acting has a negative net decision value ($W_t^{D,B}(p_0) < 0$). This implies the bad agent enjoys a positive net reputational value from acting: $W_t^{R,B}(p_0) > 0$. Moreover, if acting is reputationally beneficial at the prior, it must be even more beneficial at higher beliefs, since accuracy is rewarded in equilibrium. So, the good agent must also enjoy a positive net reputational benefit from acting at the decision-optimal cutoffs, i.e., $W_t^{R,G}(\hat{p}_t) > 0$, since these cutoffs lie well above the prior when acting is sufficiently sub-optimal ex-ante. It again follows that the good agent acts prematurely.

⁸ While this intuition applies to the case where $X \in (0, 1)$, this result holds even when $X = 1$.

5.2. Delayed action

The previous section established that reputation induces premature action when acting is highly sub-optimal ex-ante. One might expect the opposite effect when acting is highly *optimal* ex-ante, namely that the agent delays action. In this subsection, I establish that this is indeed true. However, because the agent's stopping problem is inherently one-sided (acting is irreversible, abstaining is not), the result is not symmetric to that of premature action.

Formally, *delayed action* occurs at t when $p_t^* > \hat{p}_t$, i.e., the good agent acts for a strictly smaller range of beliefs than the decision-optimal rule. I now establish that when acting is sufficiently optimal ex-ante, the agent acts with delay for at least some time. When the agent's reputational motive is sufficiently large, then they act with delay in all periods. This is formalized as [Theorem 2](#).

Theorem 2 (Delayed action). *If $p_0 > \hat{p}_t$ for all t , then action is delayed in at least some t for any $X > 0$, and action is delayed in all t if X is sufficiently large.*

This proposition establishes that the agent acts with delay for at least some periods when the prior exceeds the decision-optimal cutoffs in all t . While this may seem to establish only that action is delayed for a sufficiently high prior, it also implies that holding fixed a prior, action is delayed whenever K_1 is sufficiently large, or K_0 is insignificant. This is because, per [Proposition 2](#), the decision-optimal cutoffs are decreasing in both payoff parameters K_1 and K_0 .

The basic intuition is analogous to the case of premature action. When acting is highly optimal ex-ante, delaying action (or never acting) is costly for the bad agent's decision payoff. So in equilibrium, abstaining is a costly signal that the agent is good and a reputation-driven agent responds to this by delaying action.

To more precisely convey its reasoning, it is again helpful to separately consider the cases of a small and large reputational motive. Regardless of X , when acting is highly optimal ex-ante, abstaining in any period is damaging to the bad agent's decision value. When X is large, the bad agent nonetheless mixes in every period, as was true under a pure reputational motive. To ensure indifference, the bad agent must benefit reputationally from abstaining every period. This in turn implies an expected reputational benefit from abstaining at the good agent's decision-optimal

cutoffs $\hat{p}_t$, leading to delayed action in every period. When X is small, the damage to the bad agent’s decision value from abstaining can outweigh any reputational benefit from doing so, leading him to act with certainty by some $s \leq T$. In all periods $t \leq s$, abstaining remains reputationally beneficial, leading to delayed action. However, an agent who continues past s will have revealed that he is good, and thus enjoys a perfect reputation regardless of his behavior thereafter. So from this time onwards, he will not distort his behavior and instead employs the decision-optimal cutoffs $\hat{p}_t$. It is for this reason that [Theorem 2](#) includes the caveat that action may not be delayed in all periods when X is small.

Discussion Let us now interpret the previous two results in the context of irreversible investment. [Theorem 1](#) suggests that for projects with a low ex-ante value —because success is unlikely, the downside potential is high, or the upside potential is modest— reputational motives lead to premature investment. This implies over-investment in “long-shot” projects since investment, and especially early investment, in such projects serves as a costly signal of learning ability. Moreover, because the agent invests prematurely in every period, he also engages in an inefficiently low level of experimentation. Meanwhile, [Theorem 2](#) suggests that for projects with a high ex-ante value, the opposite phenomenon arises: investment is delayed. This implies that the agent will both underinvest and experiment for too long in promising projects, only waiting until he is extremely sure that the project will succeed before investment.

Together, these results imply that reputational motives distort the research process by either shading up or shading down the agent’s option value from learning. Furthermore, the agent distorts in a way that is opposed to the perceived profitability of a project. That is, reputation incentivizes behavior that is diametrically opposed to the optimal behavior of a naive (uninformed) investor.

5.3. Dynamics in distortions

[Theorem 1](#) and [Theorem 2](#) showed that when action is highly disadvantageous (advantageous) ex-ante, reputation induces distortions that are time-invariant in direction: the agent either acts prematurely in every period or with delay in every period. Here, I show that for decision problems where the value of acting takes an

intermediate value, the direction of distortions can dynamically reverse over time. Specifically, the agent may act prematurely in early periods, and with delay in later periods. This phenomenon arises whenever the agent is sufficiently reputationally motivated and discounting is slow (i.e., β is high). This result is formalized as [Theorem 3](#).

Theorem 3. *Suppose $p_0 \in (\hat{p}_1, \hat{p}_T)$. Fixing all other parameters, there exist $\bar{X} \in (0, 1)$ and $\bar{\beta} \in (0, 1)$ such that if $\beta > \bar{\beta}$ and $X > \bar{X}$ the agent:*

- *acts prematurely in early periods (i.e., there exists $\underline{t}$ such that $p_t^* < \hat{p}_t$ for all $t \leq \underline{t}$),*
- *acts with delay in later periods (i.e., there exists $\bar{t}$ such that $p_t^* > \hat{p}_t$ for all $t \geq \bar{t}$).*

This can be understood by considering how the decision value and reputational value of acting interact over time. Specifically, in early periods, the decision value of acting is low: discounting is minimal and there is a high option value of waiting from learning in future periods. Accordingly, the decision-optimal rule entails acting early only under very high beliefs that $\theta = 1$. However, as the deadline T approaches, this option value falls because there is little time left to learn, and so the decision-optimal cutoff falls sharply. Meanwhile, the reputation payoff evolves differently. There is always a reputational benefit to speed, and thus cost to delay, when $\theta = 1$. Importantly, this benefit from speed is sizeable in early periods even when discounting is slow. Because the decision-optimal cutoffs are high in these early periods, speed is reputationally beneficial in expectations at these cutoffs, inducing premature action. As the deadline approaches and the decision-optimal cutoff falls, speed is no longer expected to be rewarded at these cutoffs. Instead, delay becomes increasingly reputationally beneficial, as abstaining forever is costly for the bad agent's decision payoff (since $p_0 > \hat{p}_T$) and serves as a credible signal of learning ability. This induces a shift to delayed action in later periods.

Discussion This result illustrates that reputational motives can induce distortions that reverse over the course of the research process. More precisely, [Theorem 3](#) indicates that when discounting is sufficiently slow, the agent is initially too eager to act on limited information. I.e., he under-experiments. However, as he approaches the end of the research process, he becomes overly conservative, choosing to either continue experimenting excessively or to avoid acting altogether

even though doing so is not decision-optimal. This reversal arises because the reputational benefit of speed incentivizes early action, but the reputational cost from late action incentivizes conservatism late in the research process. This result bears resemblance to the finding of [Prendergast and Stole \(1996\)](#) that reputation leads managers to over-react to new information in early periods and under-react to it later. In contrast, the reversal in [Theorem 3](#) operates on a different dimension of an agent's behavior: here, reputational incentives do not distort the agent's responsiveness to new information, but rather his willingness to continue experimenting.

6. Extensions

I now present two extensions of the baseline model, allowing for: (1) learning by the bad type and (2) an infinite horizon.

6.1. Learning by the bad type

Here, I extend the model to allow for learning by the bad type. Otherwise, I maintain all assumptions of the baseline model. Formally, let us now suppose that B observes some signal s_t^B in every period t , distributed according to conditional density $f^B(\cdot|\theta)$. As with G , I suppose that B 's signals are i.i.d. given θ and are labeled as their likelihood ratios: $s_t^B \equiv \frac{f^B(s_t^B|\theta=1)}{f^B(s_t^B|\theta=0)}$. I now impose two assumptions to formalize the notion that, while B has access to an informative signal, G has a greater ability to learn. First, I assume that B 's signals have bounded support on the likelihood ratios: there exists $\underline{s} < \bar{s} \in (0, \infty)$ such that the support of f^B is $(\underline{s}, \bar{s})$. This stands in contrast to G 's signal, which is full support on the likelihood ratios. Second, I assume G 's signal is Blackwell more informative than B 's signal.

I first show that when B has access to such a signal, he does not mix but rather plays a cutoff rule in equilibrium. [Proposition 6](#) characterizes this rule.

Proposition 6. *In equilibrium, B and G play cutoff strategies $(p_t^{*,B})_{t=1}^T$ and $(p_t^{*,G})_{t=1}^T$, respectively, where $p_t^{*,B} < p_t^{*,G}$ for all $t < T$, and $p_T^{*,B} = p_T^{*,G}$.*

[Proposition 6](#) establishes that B acts for a strictly larger range of beliefs than G in all periods prior to the deadline. This is due to the fact that G has access to

a more informative signal than B in every period, granting him a strictly higher continuation value than B in $t < T$, and thus a greater willingness to abstain.

Next, I show that the main results regarding distortions in the agent's strategies ([Theorem 1](#) and [Theorem 2](#)) continue to hold, provided the likelihood ratios of B 's signals are sufficiently bounded. That is, the above results are robust to allowing B a limited ability to learn. This result is formalized as [Proposition 7](#).

Proposition 7. *Fixing all parameters except f_B , there exists $\underline{s}$ and $\bar{s}$, where $\underline{s} < \bar{s}$, such that if $\text{supp}[f^B] \subset [\underline{s}, \bar{s}]$, then [Theorem 1](#) and [Theorem 2](#) hold.*

Both [Theorem 1](#) and [Theorem 2](#) relied on the notion that the reputation function rewards behavior which is always sub-optimal from the bad agent's perspective, but sometimes optimal given the good agent's information. In the baseline model, the bad agent's perspective was the ex-ante one. This notion extends to a setting where the bad agent observes signals of bounded informativeness, as this ensures that the bad agent's perspective never drastically departs from the ex-ante one. However, allowing the bad agent to access to signals which are arbitrarily close to those of the good agent in informativeness, the result would not extend: it is the difference in learning ability that allows for inferences to be made about the agent's type in equilibrium, which in turn induces timing distortions.

6.2. Infinite horizon

The baseline model assumed an exogenous deadline $T < \infty$. I now extend the baseline model to an infinite horizon. I show that even under an infinite horizon, the agent behaves as if there is a deadline in equilibrium.

Formally, let us now assume $T = \infty$ (i.e., $t \in \mathbb{N}^+$). For simplicity, for the purposes of this extension I restrict attention to a purely reputation-driven agent ($X = 1$). Let us otherwise maintain all assumptions of the baseline model, except for the informativeness assumption ([Assumption 1](#)). I relax this assumption because, as I now show, informativeness cannot hold in every period under an infinite horizon. This is stated as [Proposition 8](#).

Proposition 8. *When $T = \infty$, [Assumption 1](#) must fail in equilibrium.*

This follows from the fact that the bad agent's indifference between acting and abstaining, which must hold in equilibrium, cannot be satisfied indefinitely. Under

an infinite horizon, the reputational benefit of speed must vanish as $t \rightarrow \infty$ (due to the boundedness of the reputation function), leading the good agent's equilibrium cutoffs to approach 1. I.e., in the limit the good agent only acts if he is arbitrarily certain that $\theta = 1$. This in turn implies acting when $\theta = 0$ reveals that the agent is the bad type, making acting sub-optimal for the bad agent in the limit and thus leading to a failure of indifference.

Since informativeness cannot hold indefinitely, I now define a relaxed version of the informativeness assumption that accomodates the agent stopping acting in finite time. I call this relaxed assumption *T-informativeness*.

Definition 1. For any $T \in \mathbb{N}^+ \cup \infty$, an equilibrium satisfies *T-informativeness* if for any $t \leq T$, for some θ , $Pr(\tau = t | i = G, \theta, \tau \not\leq t) \neq Pr(\tau = t | i = B, \theta, \tau \not\leq t)$, and for all $t > T$ and θ , $Pr(\tau = t | i = G, \theta) = 0$.

Precisely, *T-informativeness* imposes that informativeness applies for all $t \leq T$, but that the agent ceases acting after T . One can show that there exists a *T-informative* equilibrium for any $T < \infty$. This is formalized as [Corollary 3](#).

Corollary 3. For any $T < \infty$, there exists a *T-informative equilibrium*.

Together, [Proposition 8](#) and [Corollary 3](#) imply that the deadline T of the baseline model need not be exogenously imposed: under an infinite horizon, a deadline arises endogenously. That is, any equilibrium of the infinite-horizon setting must also be an equilibrium of the finite-horizon setting, and vice versa.

7. Conclusion

This paper studies how a motive to signal learning ability can distort the timing of action. This reputational motive gives rise to contrarian behavior, driven by the endogenous role speed plays in reputation formation. In equilibrium, the agent's action may be either premature or delayed, with premature action occuring when the ex-ante value of action is low, and delayed action occuring when it is high. When the probability of success takes an intermediate value, the agent's distortionary behavior can change over time, with premature action at the beginning of the research process but excess caution towards its end. More

generally, this paper illustrates how a motive to signal reputation can distort the value of experimentation driven by the endogenous, signaling value of speed in such settings.

In this paper, I have studied the behavior of a single agent in isolation, i.e., in the absence of competition. Not only would competition give rise to new strategic motives, it would allow agents to condition their investment behavior on that of their opponents. How competition between reputation-driven agents could give rise to unique dynamics in decision behavior is a question that warrants further investigation.

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8. Appendix

Before proceeding, let us define two different conditional distributions. First, let $G_t(\cdot|p_{t-1})$ denote the good agent’s distribution of time- t beliefs given that their time $t - 1$ belief was p_{t-1} . It follows from the definition of F that on-path in any equilibrium:

$$G_t(p_t|p_{t-1}) = F\left(\left(\frac{1 - p_{t-1}}{p_{t-1}}\right)\left(\frac{p_t}{1 - p_t}\right)\right).$$

Second, let $H_t(\cdot|p_{t-1})$ denote the distribution of time- t beliefs given $\tau \notin \{1, \dots, t\}$ and that their time- $t - 1$ belief was p_{t-1} . Finally, let $H_t(\cdot)$ denote the good agent’s distribution of time- t beliefs given $\tau \notin \{1, \dots, t\}$, conditional on p_0 (namely, not conditional on the time- $t - 1$ belief). It is computed recursively as follows:

$$H_1(p_1) = H_1(p_1|p_0)$$

$$H_t(p_t) = \int_0^1 H_t(p_t|p_{t-1}) dH_{t-1}(p_{t-1}).$$

Proof of Lemma 1. Proof by induction.

Base case: $t = T$. Note that

$$V_T^G(p, \emptyset) = pR(\emptyset, 1) + (1 - p)R(\emptyset, 0),$$

which is linear in p and thus convex in p .

Induction step: Fix any $t < T$, and assume by induction that $V_{t+1}(p, \emptyset)$ is convex in p . We want to show that $V_t(p, \emptyset)$ is convex in p . This is equivalent to showing that for all $p, p' \in [0, 1]$ and $\lambda \in [0, 1]$,

$$\lambda V_t(p, \emptyset) + (1 - \lambda) V_t(p', \emptyset) \geq V_t(\bar{p}),$$

where $\bar{p} \equiv \lambda p + (1 - \lambda)p'$. To this end, fix a p, p' , and $\lambda \in [0, 1]$ and define the following binary signal $b \in \{0, 1\}$ on θ :

$$Pr(b = 1 | \theta = 1) = \frac{p\lambda}{\bar{p}}, \quad Pr(b = 1 | \theta = 0) = \frac{(1 - p)\lambda}{1 - \bar{p}},$$

Now define the following two signals σ and $\tilde{\sigma}$:

$$\sigma : \{0, 1\} \rightarrow \Delta([0, \infty]), \text{ where } \sigma(\theta) = F(\cdot | \theta)$$

$$\tilde{\sigma} : \{0, 1\} \rightarrow \Delta([0, \infty] \times \{0, 1\}), \text{ where } \tilde{\sigma}(\theta) = \tilde{F}(\cdot, \cdot | \theta),$$

and for $b^* \in \{0, 1\}$, $\tilde{F}(s, b^*) = F(s | \theta) Pr(b \leq b^* | \theta)$. Note that $\tilde{\sigma}$ Blackwell dominates σ . Now assuming that the agent has prior belief p let $G_t(\cdot | p)$ and $\tilde{G}_t(\cdot | p)$ denote the distribution of posteriors after observing σ and $\tilde{\sigma}$, respectively. It follows from the Law of Iterated Expectations that:

$$\tilde{G}_t(q | \bar{p}) = Pr(b = 1 | \bar{p}) G_t(q | p) + Pr(b = 0 | \bar{p}) G_t(q | p') = \lambda G_t(q | p) + (1 - \lambda) G_t(q | p'). \quad (4)$$

Now, note that since $V_{t+1}(p, \emptyset)$ is convex in p and $V_{t+1}(p, 1) = pR(t + 1, 1) + (1 - p)R(t + 1, 0)$ is linear in p , $V_{t+1}(p) = \max\{V_{t+1}(p, \emptyset), V_{t+1}(p, 1)\}$ is also convex in p . Thus

$$\begin{aligned} V_t(\emptyset, \bar{p}) &= \int_0^1 V_{t+1}(q) dG_t(q | \bar{p}) \\ &\leq \int_0^1 V_{t+1}(q) d\tilde{G}_t(q | \bar{p}) \\ &= \lambda \int_0^1 V_{t+1}(q) dG_t(q | p) + (1 - \lambda) \int_0^1 V_{t+1}(q) dG_t(q | p') \\ &= \lambda V_t(p, \emptyset) + (1 - \lambda) V_t(p', \emptyset), \end{aligned}$$

where the inequality follows from Blackwell's (1953) theorem, and the second equality follows from (4). $\square$

Proof of Lemma 2. Fix a t , and assume by induction that (1) holds for all $s > t$.

First, consider the case where $A_s^B = 1$ for some $s < t$. By the informativeness property, $Pr(\tau < t | i = G) < 1$. Thus $R_{t-1} = 1$. It follows that $V_s^R(act, p) = V_s^R(\emptyset, p) = 1$ for all $s \geq t$ and for all p since

$$V_t^{NR}(act, 1) - V_t^{NR}(\emptyset, 1) > 0 \text{ and } V_t^{NR}(act, 0) - V_t^{NR}(\emptyset, 0) < 0,$$

and thus (1) holds in t .

Next, suppose $A_s^B < 1$ for all $s < t$. Consider the case where $A_t^B = 1$. By the informativeness property, $Pr(\tau = t | \tau \leq t, i = G) < 1$. Thus,

$$V_t^R(p, act) < V_t^R(p, \emptyset) = 1 \text{ for all } p, \quad (5)$$

and so $V_t(0, act) < V_t(0, \emptyset)$. To show that (1) holds in t , it remains to show that $V_t(1, act) > V_t(1, \emptyset)$. Assume by contradiction $V_t(1, act) \leq V_t(1, \emptyset)$. If $t = T$, then $V_t(p_0, \emptyset) > V_t(p_0, 1)$, and thus $A_t^B = 0$. Contradiction. If $t < T$, it follows from the inductive assumption that $V_t(p = 1, \emptyset) = V_{t+1}(p = 1, act)$. So,

$$V_{t+1}(p = 1, act) \leq V_t(p = 1, act).$$

By (5) and the fact that $V_{t+1}^D(0, act) > V_t(0, act)$ that $V_{t+1}(p = 0, act) > V_t(p = 0, act)$, and thus $V_{t+1}(p_0, act) > V_t(p_0, act)$. So, B can profitably deviate by choosing $A_t^B = 0$ and $A_{t+1}^B = 1$. Contradiction.

Next, consider the case where $A_t^B = 0$. By the informativeness property, $Pr(\tau = t | \tau \leq t, i = G) > 0$. Thus, $V_t^R(p, abstain) \leq V_t^R(p, act) = 1$ for all p , and so $V_t(1, act) > V_t(1, \emptyset)$. To show (1) in t , it remains to show $V_t(0, act) < V_t(0, \emptyset)$. Assume by contradiction $V_t(0, act) \geq V_t(0, \emptyset)$. Since $V_t(p, act)$ is linear in p and by Lemma 1 $V_t(p, \emptyset)$ is convex in p ,

$$V_t(p, act) > V_t(p, \emptyset) \text{ for all } p > 0.$$

It follows that

$$V_t(p_0, act) > V_t(p_0, \emptyset) \geq V_t^B(p_0, \emptyset).$$

Thus B can profitably deviate by playing $A_t^B = 1$. Contradiction.

Finally, consider the case where $A_t^B \in (0, 1)$. I claim that $A_{t+1}^B > 0$ whenever $t < T$. Suppose not, by contradiction. By the informativeness property, $Pr(\tau = t+1 | i = G) > 0$. First, consider the case where $V_1^D(act, p_0) = K_0(1-p_0) + K_1p_0 \leq 0$. Because $A_t^B \in (0, 1)$, $V_t^R(p_0, act) < 1$ and because $A_{t+1}^B = 0$, $V_{t+1}^R(act, p_0) = 1$. Thus, $V_t(p_0, act) < V_{t+1}(p_0, act)$. It follows that $A_t^B = 0$ and $A_{t+1}^B = 1$ is a profitable deviation. Contradiction.

Next, consider the case where $V_1^D > 0$. Since $V_{t+1}^R(p_0, act) = 1$, $V_{t+1}(p_0, act) > V_\tau(p_0, act)$ for all $\tau \neq t+1$, and thus $A_{t+1}^B = 1$ is a profitable deviation. Contradiction. It follows from the above statement that B is indifferent between acting at t and $t+1$ when $t < T$. Thus,

$$V_t(p_0, act) = V_{t+1}(p_0, act) \tag{6}$$

Now, we want to show $V_t(1, act) > V_t(1, \emptyset)$. Suppose not by contradiction. First, consider the case where $V_t(1, \emptyset) \geq V_t(1, act)$. By the inductive assumption, $V_t(1, \emptyset) = V_{t+1}(1, act)$ and $V_t(1, act) \leq V_{t+1}(1, act)$. Thus, by (6), $V_t(0, act) \leq V_{t+1}(0, act)$. If $X < 1$ this implies

$$R(t, 1) < R(t+1, 1) \text{ and } R(t, 0) > R(t+1, 0),$$

which is the violation of the no-reverse property. Contradiction. Now suppose $X = 1$. First, suppose $R(t, 1) < R(t+1, 1)$. In order for (6) to be satisfied, $R(t, 0) > R(t+1, 0)$, which again implies a violation of the no-reverse signaling property. Contradiction. We can analogously show that we cannot have $R(t, 0) > R(t+1, 0)$. So, it follows that $R(t, 1) = R(t+1, 1)$ and $R(t, 0) = R(t+1, 0)$. When $t = T$, this implies a failure of informativeness. When $t < T$, by the inductive assumption,

$$V_t(\emptyset, p) > V_{t+1}(act, p) = V_t(act, p) \text{ for all } p > 0.$$

Thus, $Pr(\tau = t | i = G, \theta) = 0$ for all θ . Since $A_t^B > 0$, this implies $R(t, \theta) = 0$ for all θ , and thus $V_t(act, p) = 0$ for all p . But by informativeness, $R(s, \theta) > 0$ for some

$s > t, \theta$. Thus, B can profitably deviate by choosing $A_t^B = 0$. Contradiction. $\square$

Proof of Proposition 1 (cutoff necessity of strategies). Here, we show that in any equilibrium, for any t , G plays an interior cutoff strategy (we later prove existence of an equilibrium). To this end, fix any equilibrium. Fix a t . By Lemma 2, $V_t(1, \emptyset) > V_t(1, 1)$ and $V_t(0, \emptyset) > V_t(0, 1)$. Because $V_t(p, \emptyset)$ is convex in p (Lemma 1) and $V_t(1, p) = pR(t, 1) + (1 - p)R(t, 0)$ is linear in p , there exists a unique $p_t^* \in (0, 1)$ such that

$$V_t(p, 1) > V_t(p, \emptyset) \text{ for all } p > p_t^*$$

$$V_t(p, 1) < V_t(p, \emptyset) \text{ for all } p < p_t^*.$$

Thus, the good agent's strategy must be such that

$$A_t^G(p) = \begin{cases} 0 & \text{for all } p < p_t^* \\ 1 & \text{for all } p > p_t^*. \end{cases}$$

$\square$

We now seek to establish existence of an equilibrium. To this end, let us define the correspondence Φ as follows. First, let us define R^x . Let R^x denote the reputation function that is consistent with the strategy profile $x = (p_1^*, \dots, p_T^*, A_1^B, \dots, A_T^B) \in [0, 1]^{2T}$. Formally, whenever Bayes Rule applies, $R^x(t, \theta)$ is given by

$$R^x(t, \theta) = \frac{1}{1 + \left(\frac{1-R_{t-1}}{R_{t-1}}\right) \left(\frac{Pr(\tau=t, \theta | \tau \neq t, i=B)}{Pr(\tau=t, \theta | \tau \neq t, i=G)}\right)}, \quad (7)$$

where the probabilities, including R_{t-1} , are those that obtain given the strategy profile x . The only case in which Bayes Rule does not apply is when $p_t^* = 1$ and $A_t^B = 0$ for some t , and in this case we impose $R^x(t, \theta) = 1$ for all θ .

Now, let $V_{s-1}^{G,x}(p, (\hat{p}_t)_{t=s}^T)$ denote G 's value, under belief p at time $s - 1$, from playing cutoff strategies $(\hat{p}_t)_{t=s}^T$ in periods $s, \dots, T$, respectively, given reputation function R^x and that the agent did not act in $s - 1$. Now, define the $\Phi_s^G(x)$

recursively as follows:

$$\Phi_s^G(x) \equiv \min_{\bar{p}_s \in [0,1]} \arg \max_{\bar{p}_s \in [0,1]} [V_{s-1}^{G,x}(p_0, (\bar{p}_t)_{t=s}^T)],$$

where $\bar{p}_t \equiv \Phi_t^G(x)$ for all $t > s$. Let $\Phi^G(x) \equiv (\Phi_t^G(x))_{t=1}^T$. Note that the value could have been taken at any interior belief (not necessarily p_0) and the analysis that follows would remain unchanged.

Next, let $V_s^{B,x}((b_t)_{t=s}^T)$ denote B 's value from playing strategy $A_t^B = b_t$ for all $t \geq s$, given reputation function R^x and that the agent did not act before s . Now, define the $\Phi_s^B(x)$ recursively as follows:

$$\Phi_s^B(x) \equiv \arg \max_{b_s \in [0,1]} V_s^{B,x}((b_t)_{t=s}^T),$$

where $b_t \in \Phi_t^B(x)$ for all $t > s$. Define $\Phi^B(x) \equiv \Phi_1^B(x) \times \dots \times \Phi_T^B(x)$, and finally $\Phi(x) \equiv \Phi^G(x) \times \Phi^B(x)$.

I wish to show that any fixed point of Φ is an equilibrium. To this end, I begin by establishing two lemmas.

Lemma 4. *In any fixed point of Φ , $p_t^* < 1$ for all t .*

Proof. Fix a t . Suppose by induction $p_s^* < 1$ for all $s < t$. We want to show $p_t^* < 1$. First consider the case where $A_s^B = 1$ for some $s < t$. This implies $X < 1$, otherwise, $A_s^B = 1$ cannot be part of a fixed point. It also implies $R_{t-1} = 1$, and thus $p_t^* = \hat{p}_t \in (0, 1)$.

Next, consider the case where $A_s^B < 1$ for all $s < t$. First, suppose that $A_t^B = 0$. Then $R(t, \theta = 1) = 1$ for all θ , so $p_t^* \leq \hat{p}_t < 1$. Next, suppose that $A_t^B > 0$. This implies that $V_t(p = 1, 1) > V(p = \emptyset)$. Thus, $p_t^* < 1$.

□

We are now ready to show that any fixed point of Φ is an equilibrium. This is formalized as [Lemma 5](#).

Lemma 5. *Any fixed point x of Φ , together with R^x , is an equilibrium that satisfies [Assumption 1](#) and [Assumption 2](#).*

Proof. Fix any fixed point x of Φ . First, I show that (x, R^x) satisfies [Assumption 1](#) and [Assumption 2](#). That [Assumption 2](#) is satisfied follows from the assumption that G plays cutoff strategies in every t . To see why [Assumption 1](#) holds, note by [Lemma 4](#) that for all t , $p_t^* < 1$. Now fix a t and suppose that $A_s^B < 1$ for all $s < t$.

If $A_t^B = 0$, then

$$Pr(\tau = t | i = B, \theta, \tau \not\leq t) = 0 \text{ and } Pr(\tau = t | i = G, \theta, \tau \not\leq t) > 0 \text{ for all } \theta,$$

so [Assumption 1](#) is satisfied. If $A_t^B > 0$, then

$$Pr(\tau = t | i = B, \theta = 1, \tau \not\leq t) = Pr(\tau = t | i = G, \theta = 0, \tau \not\leq t).$$

However, since G plays a cutoff strategy,

$$Pr(\tau = t | i = G, \theta = 1, \tau \not\leq t) > Pr(\tau = t | i = G, \theta = 0, \tau \not\leq t),$$

so [Assumption 1](#) is again satisfied.

It remains to show that (x, R^x) is an equilibrium. It follows from the definition of R^x that R^x is consistent with Bayes' Rule, given x . Next, I will show that given R^x , $(p_t^*)_{t=1}^T$ and $(A_t^B)_{t=1}^T$ are optimal for G and B , respectively. Since x is a fixed point, $A_t^B \in \Phi_t^B(x)$ for all x and the optimality of A_t^B follows from the definition of Φ_t^B . Next, consider G . First, we want to show that at any t , it is optimal for G to play a cutoff strategy. Now, note that because A_t^B is optimal under this fixed point for any t , the reasoning of [Lemma 2](#) applies here. Thus, (1) holds here. Because $V_t(p, \emptyset)$ is convex in p ([Lemma 1](#)) and $V_t(1, p) = pR(t, 1) + (1 - p)R(t, 0)$ is linear in p , there exists a unique $\hat{p}_t \in (0, 1)$ for each t such that

$$V_t(p, 1) > V_t(p, \emptyset) \text{ for all } p > \hat{p}_t$$

$$V_t(p, 1) < V_t(p, \emptyset) \text{ for all } p < \hat{p}_t.$$

Thus, the best response for G in every t is to play a cutoff strategy, where the cutoff is $\hat{p}_t$. It remains to show that for all t , $\hat{p}_t = p_t^*$. Fix a t and suppose by induction that $\hat{p}_s = p_s^*$ for all $s > t$. By the definition of Φ_t^G , it follows that $\Phi_t^G(x) = \hat{p}_t$.

□

Proof of Proposition 1: equilibrium existence. I now establish existence of a fixed point to Φ . It follows from Lemma 5 that this is an equilibrium. To this end, for each $\varepsilon > 0$, I define a constrained correspondence Φ^ε and show that for some ε , there exists a fixed point of Φ^ε which is also a fixed point of Φ . I proceed in a number of steps, as outlined below.

1. **Define constrained correspondence:** For any $\varepsilon \in (0, 1 - p_0)$, let Φ^ε be identical to Φ , except that the domain and range are constrained as follows:

$$\Phi^\varepsilon : [p_0, 1 - \varepsilon]^T \times [0, 1]^T \rightarrow [p_0, 1 - \varepsilon]^T \times (2^{[0,1]})^T.$$

Now, define

$$\Phi_s^{G,\varepsilon}(x) \equiv \min_{\bar{p}_s \in [p_0, 1 - \varepsilon]} \arg \max_{\bar{p}_s \in [p_0, 1 - \varepsilon]} [V_{s-1}^{G,x}(p_0, (\bar{p}_t)_{t=s}^T)], \quad (8)$$

and let $\Phi^\varepsilon(x) \equiv \Phi^{G,\varepsilon}(x) \times \Phi^B(x)$, where $\Phi^B(x)$ is defined as before.

2. **Existence of fixed point for Φ^ε :** I now claim that for any $\varepsilon < 1 - p_0$, Φ^ε has a fixed point. To prove this, I invoke the Kakutani fixed point theorem. To this end, I show that Φ^ε satisfies the following properties:

- (a) $\Phi^\varepsilon(x)$ **is non-empty for all x .** This follows from the fact that $[p_0, 1 - \varepsilon]$ and $[0, 1]$ are compact and $R^x(\tau, \theta)$ is bounded for all (τ, θ) , implying by the Extreme Value Theorem that both $\Phi_t^B(x)$ and $\Phi_t^{G,\varepsilon}(x)$ are non-empty for all t, x .
- (b) $\Phi^\varepsilon(x)$ **is convex and closed for all x .** $\Phi_t^{G,\varepsilon}(x)$ is a singleton by definition for all x, t . Now, fix an (x, t) and consider $\Phi_t^B(x)$. Now, define $\underline{b}_t = 0$, $\bar{b}_t = 1$, $\bar{b}_s = \underline{b}_s = A_s^B$ for all $s > t$. It follows that

$$\Phi_t^B(x) = \begin{cases} 1 & \text{if } V_t^{B,x}((\underline{b}_s)_{s=t}^T) < V_t^{B,x}((\bar{b}_s)_{s=t}^T) \\ 0 & \text{if } V_t^{B,x}((\underline{b}_s)_{s=t}^T) > V_t^{B,x}((\bar{b}_s)_{s=t}^T) \\ [0, 1] & \text{if } V_t^{B,x}((\underline{b}_s)_{s=t}^T) = V_t^{B,x}((\bar{b}_s)_{s=t}^T), \end{cases} \quad (9)$$

and thus $\Phi_t^B(x)$ is convex and closed. It follows that $\Phi^\varepsilon(x)$ is also convex and closed.

(c) Φ^ε is **upper hemi-continuous (UHC)**. I will show that for all t , Φ_t^B and $\Phi_t^{G,\varepsilon}$ are UHC everywhere on the domain. It follows that their Cartesian product Φ^ε is also UHC. Fix an $x \in X$ and a t . Let us begin with Φ_t^B . Now note that because $\varepsilon > 0$, $R^x(t, \theta)$ is continuous in x , and thus both $V_t^{B,x}((b_s)_{s=t}^T)$ and $V_t^{B,x}((\bar{b}_s)_{s=t}^T)$ are continuous in x . Thus, it follows from (9) that $\Phi_t^B(x)$ is UHC at x . Next, consider $\Phi_t^{G,\varepsilon}$. It again follows from the continuity of $R^x(t, \theta)$ that $V_t^{G,x}$ is continuous in x , and thus by (8), $\Phi_t^{G,\varepsilon}(x)$ is continuous in x .

It follows then from the Kakutani fixed point theorem that Φ^ε has a fixed point.

3. **Show that for some $\varepsilon > 0$, Φ^ε has an interior fixed point:** I now claim that for some $\varepsilon > 0$, Φ^ε has a fixed point that lies within $[0, 1 - \varepsilon]^T \times [0, 1]^T$ (i.e., a fixed point such that $p_t^* < 1 - \varepsilon$ for all t). Suppose not, by contradiction. Then, there exists $t^* < T$ and a sequence $\{\varepsilon_n\}_{n=1}^\infty$ such that $\varepsilon_n > 0$ for all n , $\lim_{n \rightarrow \infty} \varepsilon_n = 0$ and there exists a sequence $\{x_n\}_{n=1}^\infty$ where x_n is a fixed point of Φ^{ε_n} such that $p_{t^*}^* = 1 - \varepsilon_n$.

I first claim that there exists an N such that if $n > N$, $A_{t^*}^B \in (0, 1)$ under x_n . Suppose not, by contradiction. First consider the case where for some infinite subsequence $\{x_m\}_{m=1}^\infty$ of $\{x_n\}_{n=1}^\infty$, $A_{t^*}^B = 0$ for all m . It follows that $R^{x_m}(t^*, \theta) = 1$ for all θ . Thus, for m sufficiently large, $p_{t^*}^* = 1 - \varepsilon_m$ cannot be optimal. Contradiction. Next, consider the case where for some infinite subsequence $\{x_m\}_{m=1}^\infty$, $A_{t^*}^B = 1$ for all m . Since for all m , $p_{t^*}^* = 1 - \varepsilon_m$ is optimal and $p_0 < p_{t^*}^*$ for m sufficiently large, it follows that for some m $A_{t^*}^B = 1$ is not optimal. Contradiction.

Hereafter, let us thus assume that the sequence $\{x_n\}_{n=1}^\infty$ is such that $A_{t^*}^B \in (0, 1)$ for all n . I now claim that

$$\lim_{n \rightarrow \infty} R^{x_n}(s, 1) - R^{x_n}(s + 1, 1) = 0 \text{ and } \lim_{n \rightarrow \infty} R^{x_n}(s, 0) = 0 \quad (10)$$

for all $s \geq t^*$. Proof by induction. Begin with $s = t^*$. Note that by the contradiction assumption, for all n , $V_{t^*}^G(1, 1 - \varepsilon_n) \leq V_{t^*}^G(\emptyset, 1 - \varepsilon_n)$ (where this is the value function that obtains from R^{x_n}) because otherwise $p_{t^*}^* < 1 - \varepsilon_n$ under x_n . I claim this implies $\lim_{n \rightarrow \infty} R^{x_n}(t^*, 1) - R^{x_n}(t^* + 1, 1) = 0$. Suppose not, by

contradiction. Then there exists $\delta > 0$ and an infinite subsequence $\{\varepsilon_{n_k}\}_{k=1}^\infty$ of $\{\varepsilon_n\}_{n=1}^\infty$ where $n_1 < n_2 < \dots \in \mathbb{N}$ such that $R^{x_{n_k}}(t^*, 1) - R^{x_{n_k}}(t^* + 1, 1) > \delta$ for all k . Thus, there exists k such that $V_{t^*}^G(1, 1 - \varepsilon_{n_k}) - V_{t^*}^G(\emptyset, 1 - \varepsilon_{n_k}) > 0$. Contradiction.

Next, I show $\lim_{n \rightarrow \infty} R^{x_n}(t^*, 0) = 0$. Recall that by Bayes Rule, under any x_n :

$$R^{x_n}(t^*, 0) = \frac{1}{1 + (\frac{1-R_0}{R_0})(\frac{1}{Q_{t^*}(n)})} \text{ where } Q_t(n) \equiv \frac{Pr(\theta = 0 | \tau = t, i = G)Pr(\tau = t | i = G)}{Pr(\theta = 0 | \tau = t, i = B)Pr(\tau = t | i = B)},$$

and the probabilities are those that obtain under the strategy profile x_n . I claim that $\lim_{n \rightarrow \infty} Q_{t^*}(n) = 0$. Suppose not, by contradiction. Since $\lim_{n \rightarrow \infty} \varepsilon_n = 0$, $\lim_{n \rightarrow \infty} \frac{Pr(\theta=0|\tau=t,i=G)}{Pr(\theta=0|\tau=t,i=B)} = 0$, and thus it suffices to show that $\frac{Pr(\tau=t^*|i=G)}{Pr(\tau=t^*|i=B)}$ does not diverge as $n \rightarrow \infty$. This is only possible if there exists a subsequence $\{\varepsilon_{n_k}\}_{k=1}^\infty$ of $\{\varepsilon_n\}_{n=1}^\infty$ such that $\lim_{k \rightarrow \infty} \frac{Pr(\tau=t^*|i=G)}{Pr(\tau=t^*|i=B)} = \infty$. This implies $\lim_{k \rightarrow \infty} R^{x_{n_k}}(t^*, \theta) = 1$ for all θ , and thus for k sufficiently large, $A_{t^*}^B = 1$ is a profitable deviation from what is specified under x_{n_k} . Thus, x_{n_k} is not a fixed point of $\Phi^{\varepsilon_{n_k}}$. Contradiction.

Now, fix some $t > t^*$ and assume by induction that (10) holds for all s such that $t^* \leq s < t$. We want to show that it also holds for t . First, let us show that $\lim_{n \rightarrow \infty} R^{x_n}(t, 0) = 0$. Since for all n , $A_t^B \in (0, 1)$:

$$p_0 R^{x_n}(t, 1) + (1 - p_0) R^{x_n}(t, 0) = p_0 R^{x_n}(t - 1, 1) + (1 - p_0) R^{x_n}(t - 1, 0).$$

Thus,

$$\lim_{n \rightarrow \infty} R^{x_n}(t, 0) = \frac{p_0}{1 - p_0} [\lim_{n \rightarrow \infty} [R^{x_n}(t - 1, 1) - R^{x_n}(t, 1)] - \lim_{n \rightarrow \infty} R^{x_n}(t - 1, 0)] = 0,$$

where the last equality follows from the inductive assumption.

Next, let us show that $\lim_{n \rightarrow \infty} R^{x_n}(t, 1) - R^{x_n}(t + 1, 1) = 0$. Suppose not, by contradiction. Then, there exists $\delta > 0$ and subsequence $\{\varepsilon_{n_k}\}_{k=1}^\infty$ of $\{\varepsilon_n\}_{n=1}^\infty$ such that $R^{x_{n_k}}(t, 1) - R^{x_{n_k}}(t + 1, 1) \geq \delta$ for all k . This implies that there exists $p \in (p_0, 1)$ such that $p_t^* \leq p$ under x_{n_k} for all k . However, $\lim_{n \rightarrow \infty} R^{x_n}(t, 0) = 0$, and thus for all $p \in (p_0, 1)$, there exists an $N \in \mathbb{N}$ such that $p_t^* > p$ under x_n for all $n > N$. Contradiction.

Now, note that for all n , $p_t^* = p_0$ under x_n (this follows from identical reasoning to that presented in the proof of [Claim 1](#)). Thus, $Q_T(n)$ does not converge to 0 as $n \rightarrow \infty$. However, because $\lim_{n \rightarrow \infty} R^x(T, 0) = 0$, $\lim_{n \rightarrow \infty} Q_T(n) = 0$. Contradiction.

4. **This interior fixed point of Φ^ε is also a fixed point of Φ :** Fix an $\varepsilon > 0$ such that there is a fixed point x of Φ^ε such that $x \in [p_0, 1 - \varepsilon)^T \times [0, 1]^T$. I claim that x is also a fixed point of Φ . This is equivalent to showing that for all t :

$$A_t^B \in \Phi_t^B(x) \text{ and } p_t^* = \Phi_t^G(x).$$

Note that $A_t^B \in \Phi_t^B(x)$ holds because this is necessary for x to be a fixed point of Φ^ε . Next, let us show that $p_t^* = \Phi_t^G(x)$ for all t . Proof by induction. Fix a t , and suppose $p_s^* = \Phi_s^G(x)$ for all $s > t$. We want to show $p_t^* = \Phi_t^G(x)$.

By the same reasoning that is presented in the proof of [Proposition 1](#), since $p_t^* < 1 - \varepsilon$,

$$V_t^G(p, 1) > V_t^G(p, \emptyset) \text{ for all } p > 1 - \varepsilon.$$

where this is the value function that obtains given the reputation function R^x . Thus,

$$V_t^{G,x}(p_0, (p_s^*)_{s=t}^T) > V_t^{G,x}(p_0, (\tilde{p}_s)_{s=t}^T)$$

for any $\tilde{p}_t > 1 - \varepsilon$ and $\tilde{p}_s = p_s^*$ for all $s > t$. This, combined with the fact that $p_t^* = \Phi_t^{G,\varepsilon}(x)$, implies $p_t^* = \Phi_t^G(x)$.

□

Proof of [Proposition 2](#). Part 1 follows from the fact that the B agent holds belief p_0 in every period. For part 2, let us first show that G plays a cutoff rule and these cutoffs are unique. Proof by induction.

Base case: In period T , the agent acts if and only if his belief lies above $p^* \equiv \frac{-K_0}{K_1 - K_0}$.

Induction step: Fix a t and suppose that the agent plays an interior cutoff rule in all $s < t$. Note that

$$V_t(1, p) = \beta^t(pK_1 + (1 - p)K_0).$$

Now, let us observe three facts about $V_t^G(\emptyset, p)$:

1. Since a cutoff rule is played in $t + 1$,

$$V_t^G(\emptyset, 1) = V_{t+1}^G(1, 1) = \beta^{t+1} K_1 < \beta^t K_1 = V_t^G(1, 1)$$

2. $V_t^G(\emptyset, 0) = 0 > \beta^t K_0 = V_t^G(1, 0)$

3. $V_t(\emptyset, p)$ is convex in p .

These three facts together with the linearity of $V_t(1, p)$ imply that there is a unique $p_t^* \in (0, 1)$ such that (2) holds.

Now, it remains to show that the p_t^* are strictly decreasing in t . To this end, fix a $t < T$. Suppose by contradiction that $p_t^* \leq p_{t+1}^*$. Then

$$V_t^G(1, p_t^*) = V_t^G(\emptyset, p_t^*)$$

$$V_{t+1}^G(1, p_t^*) \leq V_{t+1}^G(\emptyset, p_t^*)$$

Since all these values are strictly positive

$$\beta = \frac{V_{t+1}^G(1, p_t^*)}{V_t^G(1, p_t^*)} \leq \frac{V_{t+1}^G(\emptyset, p_t^*)}{V_t^G(\emptyset, p_t^*)}. \quad (11)$$

Now, let $\tilde{V}_t(\emptyset, p)$ denote the agent's value from the modified problem which is identical to the original problem except that the time horizon is $T - 1$. It follows that for all $t < T$:

1. $\tilde{V}_t^G(\emptyset, p) = \frac{V_{t+1}^G(\emptyset, p)}{\beta}$
2. $\tilde{V}_t^G(\emptyset, p) < V_t^G(\emptyset, p)$.

These two facts together imply

$$\frac{V_{t+1}^G(\emptyset, p_t^*)}{V_t^G(\emptyset, p_t^*)} < \beta,$$

contradicting (11). □

Before proceeding, let us define the agent's *interim reputation* as follows:

Definition 2 (Interim reputation). The agent's time t interim reputation is the

principal's belief $i = G$ given that they did not report at or before t :

$$R_t \equiv \Pr(i = G | \tau \notin \{1, \dots, t\}).$$

Lemma 6. *In any equilibrium, if for all $s \leq t$ there exists a $p_s^* \in (0, 1)$ such that*

$$A_t^G(p) = \begin{cases} 0 & \text{for all } p < p_t^* \\ 1 & \text{for all } p > p_t^* \end{cases}$$

and $A_t^B \in (0, 1)$, then $R_t \in (0, 1)$.

Proof. Fix a t , and assume A^G and A^B satisfy the assumptions specified in Lemma 6. We want to show that $R_t \in (0, 1)$. Proof by induction.

Base case: $s = 0$. $R_s = R_0 \in (0, 1)$ by assumption.

Induction step: For any $s \in \{1, \dots, t\}$, assume $R_{s-1} \in (0, 1)$. We want to show that $R_s \in (0, 1)$. It follows from Bayes Rule that

$$R_s = \frac{1}{1 + \frac{\Pr(\tau \neq s | \tau \notin \{1, \dots, s_{t-1}\}, i=B)}{\Pr(\tau \neq s | \tau \notin \{1, \dots, s_{t-1}\}, i=G)}}. \quad (12)$$

To show that $R_s \in (0, 1)$, it suffices to show that both the conditional probabilities in (12) lie in $(0, 1)$. In equilibrium,

$$\Pr(\tau \neq s | \tau \notin \{1, \dots, s_{t-1}\}, i = B) = A_t^B \in (0, 1),$$

where $A_t^B \in (0, 1)$ holds by assumption. It remains to show that $\Pr(\tau \neq s | \tau \notin \{1, \dots, s_{t-1}\}, i = G) \in (0, 1)$. To this end, because the good agent is playing a cutoff strategy,

$$H_t(p_t | p_{t-1}) = \begin{cases} 0 & \text{for all } p < p_t^* \\ \frac{G_t(p_t | p_{t-1}) - G_t(p_t^* | p_{t-1})}{G_t(p_t^* | p_{t-1})} & \text{for all } p > p_t^* \end{cases}$$

We can write

$$\Pr(\tau \neq s | \tau \notin \{1, \dots, s-1\}, i = G) = \int_0^1 G_t(p_t^* | p_{t-1}) dH_{t-1}(p_{t-1}). \quad (13)$$

Now, we make two observations:

1. $G_t(p_t^*|p_{t-1}) \in (0, 1)$ for all $p_{t-1} \in (0, 1)$.
2. $H_{t-1}(p_{t-1})$ is continuous in p_{t-1} , following from the continuity of $G_{t-1}(p_{t-1}|p_{t-2})$ in p_{t-1} .

It follows from the above two observations, combined with (13) that $Pr(\tau \neq s | \tau \notin \{1, \dots, s-1\}, i = G) \in (0, 1)$.

□

Proof of Proposition 3. First, we want to show that in any equilibrium $A_t^B \in (0, 1)$ for all t . Proof by induction. Assume by induction that $A_s^B \in (0, 1)$ for all $s < t$ (this holds vacuously when $t = 1$). Assume by contradiction $A_t^B \in \{0, 1\}$. First, consider the case where $A_t^B = 0$. It follows from Bayes Rule that

$$R(t, 0) = \frac{1}{1 + \left(\frac{1-R_{t-1}}{R_{t-1}}\right) \left(\frac{Pr(\tau=t, \theta=0 | \tau \notin \{1, \dots, t-1\}, i=B)}{Pr(\tau=t, \theta=0 | \tau \notin \{1, \dots, t-1\}, i=G)}\right)} \quad (14)$$

First, note that

$$Pr(\tau = t, \theta = 0 | \tau \notin \{1, \dots, t-1\}, i = B) = A_t^B = 0.$$

Meanwhile,

$$Pr(\tau = t, \theta = 0 | \tau \notin \{1, \dots, t-1\}, i = G) = \int_0^1 \int_{p_t^*}^1 (1-p_t) dG_t(p_t|p_{t-1}) dH_{t-1}(p_{t-1}) > 0,$$

where the strict inequality follows from the fact that $p_t^* \in (0, 1)$. By Lemma 6, it follows from (14) that $R(t, 0) = 1$. One can analogously show that $R(t, 1) = 1$. Thus,

$$V_t(p_0, 1) = p_0 R(t, 1) + (1 - p_0) R(t, 0) = 1 \quad (15)$$

Now, by the Law of Iterated Expectations

$$\begin{aligned} R_{t-1} &= Pr(i = G, \tau = t | \tau \notin \{1, \dots, t-1\}) (1) \\ &\quad + Pr(i = G, \tau \neq t | \tau \notin \{1, \dots, t-1\}) \int_0^1 V_t^G(\emptyset, p) dH_t(p_t) \\ &\quad + Pr(i = B | \tau \notin \{1, \dots, t-1\}) V_t^B(p_0, \emptyset). \end{aligned} \quad (16)$$

Because R is consistent with the A^i in equilibrium, $V_t^B(p_0, \emptyset) \geq \int_0^1 V_t^G(p, \emptyset) dH_t(p_t)$. Because $R_{t-1} < 1$ ([Lemma 6](#)), it follows from (16) that $V_t^B(\emptyset, p_0) < 1$. Combining this with (15) implies $V_t^B(p_0, \emptyset) < V_t(p_0, 1)$. Thus, $A_t^B(p_0) = 1$. Contradiction.

Now, we want to show that in any equilibrium, $p_T^* = p_0$. Fix any equilibrium. We have already shown $A_t^B \in (0, 1)$. Thus,

$$V_T(p_0, 1) = V_T^B(p_0, \emptyset) = V_T^G(p_0, \emptyset), \quad (17)$$

Note further that (1) both $V_T(p, 1)$ and $V_T(p, \emptyset)$ are linear in p and (2) by [Lemma 2](#), $V_T(0, 1) < V_T^G(0, \emptyset)$ and $V_T(1, 1) > V_T^G(1, \emptyset)$. These two facts, combined with (17), imply that $V_T(p, 1) < V_T(p, \emptyset)$ for all $p < p_0$ and $V_T(p, 1) > V_T(p, \emptyset)$ for all $p > p_0$. Thus $p_T^* = p_0$.

It remains to show that for all $t < T$, $p_t^* > p_0$. To this end, fix a $t < T$. It follows from the above that B mixes between $a \in \{1, \emptyset\}$ in every t , and thus

$$V_t(p_0, 1) = V_t^B(p_0, \emptyset) \quad \text{and} \quad V_t(p_0, 1) = V_t^B(p_0, \emptyset).$$

So, $V_t(p_0, 1) = V_{t+1}(p_0, 1)$. Now it follows from [Lemma 2](#) that

$$V_t(1, \emptyset) = V_{t+1}(1, 1) \quad \text{and} \quad V_t(0, \emptyset) > V_{t+1}(0, 1).$$

By [Lemma 1](#), it follows that $V_t(p, \emptyset) > V_{t+1}(p, 1)$ for all $p < 1$. Because $p_0 \in (0, 1)$, then

$$V_t(p_0, \emptyset) > V_{t+1}(p_0, 1). \quad (18)$$

Finally, it follows from the same reasoning presented in the proof of [Proposition 1](#) that $V_t^G(p, \emptyset) > V_t(p, 1)$ if and only if $p < p_t^*$. Thus, it follows from (18) that $p_t^* > p_0$.

□

Proof of Claim 1. That $A^B \in (0, 1)$ and $p^* = p_0$ follows directly from [Proposition 3](#). It remains to show that there is a unique equilibrium. Namely, we want to show that there exists a unique $b \in (0, 1)$ such that $(A^B = b, p^* = p_0)$ is an equilibrium strategy. First, define

$$W(a, b) \equiv p_0 R^b(a, 1) + (1 - p_0) R^b(a, 0)$$

where

$$R^b(a, \theta) \equiv \frac{1}{1 + \frac{1-R_0}{R_0} \frac{Pr(a, \theta | i=B, A^B=b)}{Pr(a, \theta | i=G, p^*=p_0)}}$$

is the unique reputation function that is consistent with the strategy profile $(A^B = b, p^* = p_0)$. I claim that there exists a unique $b \in (0, 1)$ such that $W(1, b) = W(\emptyset, b)$. First, note that

$$Pr(a, \theta | i = G, p^* = p_0) \in (0, 1) \text{ for all } a, \theta. \quad (19)$$

Now, I make two observations about W :

1. $W(1, b = 0) - W(\emptyset, b = 0) > 0$ and $W(1, b = 1) - W(\emptyset, b = 1) < 0$.

To show this, note that $Pr(1, \theta | i = B, A^B = 0) = 0$ for all θ . Thus, by (19), $R^{b=0}(1, \theta) = 1$ and $R^{b=0}(\emptyset, \theta) < 1$ for all θ . Thus, $W(1, b = 0) - W(\emptyset, b = 0) > 0$. One can analogously show that $W(1, b = 1) - W(\emptyset, b = 1) < 0$.

2. $W(1, b) - W(\emptyset, b)$ is continuous and strictly decreasing in b .

To show this, note that

$$R^b(1, 1) = \frac{1}{1 + \frac{1-R_0}{R_0} \frac{p_0 b}{1-F(1|\theta=1)}},$$

which is continuous and strictly decreasing in b . One can similarly show that $R^b(1, \theta)$ ($R^b(\emptyset, \theta)$) is continuous and strictly decreasing (increasing) in b for all θ . The statement then follows from the definition of W .

1. and 2. above imply that there exists a unique b such that $W(1, b) = W(\emptyset, b)$.

Finally, I claim that $(A^B = b, p^* = p_0)$ is the unique equilibrium strategy profile. Because $W(1, b) = W(\emptyset, b)$, $V(p_0, 1) = V(p_0, \emptyset)$ and thus $A^B = b$ is a best response. That $p^* = p_0$ is a best response follows from the fact that $V(p, 1) - V(p, \emptyset)$ is strictly increasing in p . Thus, we have shown $(A^B = b, p^* = p_0)$ is an equilibrium. It remains to show uniqueness. This follows from the fact that b is the unique value such that $W(1, b) = W(\emptyset, b)$, and thus the unique value such that $V(p_0, 1) = V(p_0, \emptyset)$ under the R that is consistent with this A^B . $\square$

Before proceeding, I state and establish a useful Lemma: when $X = 1$, for any given belief $p \in (0, 1)$, the agent's equilibrium cutoffs have to lie below p if the prior is sufficiently low.

Lemma 7. Assume $X = 1$, and fix all parameters except p_0 . For any $p \in (0, 1)$, there exists $\bar{p} \in (0, 1)$ such that if $p_0 < \bar{p}$, $p_t^* < \bar{p}$ in any equilibrium under p_0 .

Proof. Fix any $p \in (0, 1)$. To prove the statement, it suffices to show that for each t , there exists $\bar{p}_t \in (0, 1)$ such that if $p_0 < \bar{p}_t$, then $p_t^* < p$ for any equilibrium under p_0 .

Suppose not, by contradiction. Let s be the last period such that the statement fails. Since by [Proposition 3](#) $p_T^* = p_0$ in any equilibrium, $s < T$. Thus, there exists a sequence of priors $(p_0^n)_{n=1}^\infty$ such that $\lim_{n \rightarrow \infty} p_0^n = 0$, where

$$p_s^{*,n} > q \text{ for all } n \text{ and } \lim_{n \rightarrow \infty} p_{s+1}^{*,n} = 0, \quad (20)$$

where $(p_t^{*,n})_{n=1}^\infty$ is some equilibrium under p_0^n . Let superscript n refer to objects under this equilibrium.

It follows from B 's indifference ([Proposition 3](#)) that

$$\lim_{n \rightarrow \infty} [R^n(s+1, \theta=0) - R^n(s, \theta=0)] = 0. \quad (21)$$

Thus, in order for the $p_s^{*,n}$ to be optimal for all n

$$\lim_{n \rightarrow \infty} [R^n(s+1, \theta=1) - R^n(s, \theta=1)] = 0. \quad (22)$$

Now, it follows from Bayes Rule that for any t ,

$$R^n(t, \theta) = \frac{1}{1 + \frac{1-R_{t-1}^n}{R_{t-1}^n} X(t, \theta)},$$

where

$$X^n(t, \theta) \equiv \frac{A_t^{B,n} p_0^n}{Pr(\tau = t | \tau \not\prec t, i = G) Pr(\theta | \tau = t, i = G)}.$$

Thus, in order for (21) and (22) to hold,

$$\lim_{n \rightarrow \infty} X^n(s, \theta) = \lim_{n \rightarrow \infty} X(s+1, \theta) \text{ for } \theta \in \{0, 1\}. \quad (23)$$

However, it follows from (20) that

$$\lim_{n \rightarrow \infty} \frac{Pr(\theta = 1 | \tau = s | i = G)}{Pr(\theta = 1 | \tau = s + 1, i = G)} = \infty \text{ while } \lim_{n \rightarrow \infty} \frac{Pr(\theta = 0 | \tau = s | i = G)}{Pr(\theta = 0 | \tau = s + 1, i = G)} = 0,$$

which implies that (23) fails. Contradiction. $\square$

Proof of Proposition 4. Fix any R_0 , T and f . Consider some $p_0 \in (0, 1)$. By Lemma 7, there exists a $\bar{p} \in (0, p_0)$ such that if $p'_0 < \bar{p}$, $p_t^* < \bar{p}$ in any equilibrium strategy $(p_t^*)_{t=1}^T$ under p'_0 , for all t . Furthermore, it follows from Proposition 3 that $p_r^* \geq p_0$ for all t in any equilibrium strategy $(p_t^*)_{t=1}^T$ under p_0 . Thus, $p_t^* < p_t^*$ for all t , under any such equilibria. $\square$

Proof of Proposition 5. Fix any $t < T$. We want to show that

$$R(t, \theta = 1) > R(t + 1, \theta = 1) \text{ and } R(t, \theta = 0) < R(t + 1, \theta = 0).$$

First, note that $G(p_{t+1} | p_t = 1) = F(0) = 1$. Thus, since $p_{t+1}^* \in (0, 1)$,

$$V_t(1, \emptyset) = \int_0^{p_{t+1}^*} V_{t+1}(p_{t+1}, \emptyset) dG_t(p_{t+1} | p_t = 1) + \int_{p_{t+1}^*}^1 V_{t+1}(p_{t+1}, 1) dG_t(p_{t+1} | p_t = 1) = V_{t+1}(1, 1).$$

So,

$$V_t(1, 1) = V_{t+1}(1, 1) \Leftrightarrow R(t, 1) = R(t + 1, 1). \quad (24)$$

Next, recall that by the bad agent's indifference condition,

$$V_t(1, p_0) = V_{t+1}(1, p_0) \Leftrightarrow p_0 R(t, 1) + (1 - p_0) R(t, \theta = 0) = p_0 R(t + 1, 1) + (1 - p_0) R(t + 1, 0)$$

This, combined with (24), implies that $R(t, 0) < R(t + 1, 0)$. $\square$

Proof of Lemma 3. Because G plays a cutoff strategy, it suffices to show that

$V_t^G(1, \hat{p}_t) \geq V_t^G(\emptyset, \hat{p}_t)$. Note that

$$V_t^{NR,G}(\hat{p}_t, 1) = \hat{V}_t^G(\hat{p}_t, 1) = \hat{V}_t^G(\hat{p}_t, \emptyset) \geq V_t^{NR,G}(\hat{p}_t, \emptyset),$$

where the second equality follows from the definition of $\hat{p}_t$. Thus,

$$V_t^G(1, \hat{p}_t) = (1-X)V_t^{NR,G}(1, \hat{p}_t) + XV_t^R(1, \hat{p}_t) > (1-X)V_t^{NR,G}(\emptyset, \hat{p}_t) + XV_t^R(\emptyset, \hat{p}_t) = V_t^G(\emptyset, \hat{p}_t),$$

where the strict inequality follows from the assumption that $V_t^R(1, \hat{p}_t) > V_t^R(\emptyset, \hat{p}_t)$. $\square$

Proof of Theorem 1. First, let us consider K_0 and K_1 . Begin by considering the case where $X < 1$. Fixing all other parameters, we want to show that there exists $\bar{K} > 0$ such that if $K_1 < \bar{K}$, then $p_t^* < \hat{p}_t$ for all t under any equilibrium (the statement for K_0 follows analogously). Suppose not, by contradiction. Then, there exists a strictly decreasing sequence of K_1 , $\{K_1^n\}_{n=1}^\infty$ such that $K_1^n > 0$ for all n and $\lim_{n \rightarrow \infty} K_1^n = 0$ such that there exists an equilibrium strategy $\{p_t^{*,n}\}_{t=1}^T$ associated with each K_1^n where $p_s^{*,n} \leq \hat{p}_s^n$ for all n and some $s < T$ where $\hat{p}_s^n$ is the decision-optimal cutoff under K_1^n . In what follows, let superscript n denote objects for said equilibrium under K_1^n .

Since $\lim_{n \rightarrow \infty} \hat{p}_s^n = 1$, it must be that $\lim_{n \rightarrow \infty} p_s^{*,n} = 1$. In order for such $p_s^{*,n}$ to be optimal, it must be that:

$$\lim_{n \rightarrow \infty} [R^n(s, \theta = 1) - R^n(s + 1, \theta = 1)] = 0 \quad (25)$$

Meanwhile, since we cannot have $R^n(s, \theta = 0) = R^n(s, \theta = 1) = 1$, $A_s^{B,n} \in (0, 1)$ for all n . Now note there exists an N and $\bar{W} < 0$ such that for all $n > N$, $W_t^{D,B,n}(p_0) < \bar{W}$. Thus, to maintain B 's indifference, for all $n \geq N$, $W_t^{R,B,n}(p_0) > -\bar{W} > 0$, where:

$$\begin{aligned} W_t^{R,B,n}(p_0) &= [R^n(s, 1) - R^n(s + 1, 1)]p_0 + [R^n(s, 0) - R^n(s + 1, 0)](1 - p_0) \\ &< R^n(s, 1) - R^n(s + 1, 1). \end{aligned}$$

Thus, for all $n \geq N$, $R^n(s, 1) - R^n(s + 1, 1) > -\bar{W} > 0$, which contradicts (25).

Next, suppose $X = 1$. I claim that for all t , there exists $\bar{p}_t \in (0, 1)$ such that $p_t^* < \bar{p}_t$. This proves the statement for K_0 and K_1 under $X = 1$. Suppose not, by

contradiction. Let s be the last period where the statement fails. Since $p_T^* = p_0$ in any equilibrium, $s < T$. Thus, there exists a sequence of equilibria with time- s strategies $p_s^{*,n}$ such that the $p_s^{*,n}$ are strictly increasing in n and $\lim_{n \rightarrow \infty} p_s^{*,n} = 1$. This implies

$$\lim_{n \rightarrow \infty} [R^n(s, 1) - R(s + 1, 1)] = 0,$$

which implies

$$\lim_{n \rightarrow \infty} [R^n(s, 0) - R(s + 1, 0)] = 0.$$

Meanwhile, $\lim_{n \rightarrow \infty} R^n(s, 0) = 0$, because otherwise $\lim_{n \rightarrow \infty} R^n(s, 1) = 1$, which would imply that for n sufficiently large, $p_s^{*,n}$ is not optimal. Thus, $\lim_{n \rightarrow \infty} R^n(s + 1, 0) = 0$. However, for all n ,

$$R^n(s + 1, 0) = \frac{1}{1 + \frac{1-R_0}{R_0} \frac{Pr^n(\tau=s+1|i=B)}{Pr^n(\tau=s+1|i=G, \theta=0)}}.$$

Since there exists $\bar{p}_{s+1} \in (0, 1)$ such that $p_{s+1}^* < \bar{p}_s$, there exists a $Q > 0$ such that $p_{s+1}^* < \bar{p}_s$, there exists a $Q > 0$ such that $Pr^n(\tau = s + 1 | i = G, \theta = 0) > Q$. Thus, we cannot have $\lim_{n \rightarrow \infty} R^n(s + 1, 0) = 0$. Contradiction.

Next, consider p_0 . Fix all parameters except p_0 and first assume $X \in (0, 1)$. I show that there exists $\bar{p} \in (0, 1)$ such that if $p_0 < \bar{p}$, $p_t^* < \hat{p}_t$ for all $t \in \{1, \dots, T\}$ in any equilibrium. Fix any $t \in \{1, \dots, T\}$. I will show there exists $\bar{p}_t \in (0, 1)$ such that if $p_0 < \bar{p}_t$, then $p_t^* < \hat{p}_t$ under any equilibrium strategy of G . Letting $\bar{p} \equiv \min_t \bar{p}_t$, this implies the statement we wish to prove.

Suppose by contradiction that for some t , there does not exist such a $\bar{p}_t$. Then, there exists a decreasing sequence $\{p_{0,n}\}_{n=1}^\infty$ such that $p_{0,n} \in (0, \hat{p})$ for all n and $\lim_{n \rightarrow \infty} p_{0,n} = 0$, where for all n and for some equilibrium strategy profile $(A_{t,n}^B, p_{t,n}^*)_{t=1}^T$ under prior $p_{0,n}$, $p_{t,n}^* \geq \hat{p}_t$. Now, I proceed in a number of steps:

1. **Show that for all n , $A_{s,n}^B > 0$ for all $s \geq t$.** Begin with time t . Suppose by contradiction that $A_{t,n}^B = 0$ for some n . Because $p_{t,n}^* \geq \hat{p}_t > 0$, $R_n(t, \theta) = 1$ for all θ , where R_n is the reputation function associated with equilibrium $(A_{t,n}^B, p_{t,n}^*)_{t=1}^T$. Thus, $V_{t,n}^{R,G} = 1$, where $V_{t,n}^{R,G}$ is G 's reputational value under the equilibrium. Now, let $R_{t-1,n}$ denote the interim reputation under the equilibrium. Because $A_{s,n}^B < 1$ for all $s < t$, it follows that $R_{t-1,n} < 1$. Thus, it must be that $V_{t,n}^{R,G}(\hat{p}_t, \emptyset) < 1$. Thus by [Lemma 3](#), $p_{t,n}^* < \hat{p}_t$. Contradiction.

Next, consider some $s > t$. Suppose by contradiction that $A_{s,n}^B = 0$ for some n . By Lemma 2, $p_{s,n}^* < 1$, and thus by Bayes Rule $R_n(s, \theta) = 1$. But then:

$$\begin{aligned} V_{s,n}^B(p_{0,n}, 1) &= (1 - X)(p_{0,n}K_1 + (1 - p_{0,n}K_0)\beta^s + X \\ &> (1 - X)(p_{0,n}K_1 + (1 - p_{0,n}K_0)\beta^t + XV_{t,n}^R(p, 1) = V_{t,n}^B(1, p_{0,n}), \end{aligned} \quad (26)$$

where the strict inequality follows from the fact that $p_{0,n} < \hat{p}$ by assumption, and thus $p_{0,n}K_1 + (1 - p_{0,n})K_0 < 0$.

2. **Show** $\lim_{n \rightarrow \infty} [\mathbf{R}_n(s, 0) - \mathbf{R}_n(s + 1, 0)] > 0$ for all s such that $t \leq s < T$, and $\lim_{n \rightarrow \infty} [\mathbf{R}_n(T, 0) - \mathbf{R}_n(\emptyset, 0)] > 0$.

Fix some s such that $t \leq s < T$, By 1., $V_{s,n}^B(1, p_{0,n}) = V_{s+1,n}^B(1, p_{0,n})$ for all n , where

$$V_{n,s}^B(1, p_{0,n}) = (1 - X)[p_{0,n}K_1 + (1 - p_{0,n})K_0]\beta^s + X[p_{0,n}R_n(s, 1) + (1 - p_{0,n})R_n(s, 0)].$$

Thus,

$$\lim_{n \rightarrow \infty} V_{s,n}^B(1, p_{0,n}) - V_{s+1,n}^B(1, p_{0,n}) = (1 - X)\beta K_0 + X \lim_{n \rightarrow \infty} [R_n(s, 0) - R_n(s + 1, 0)] = 0.$$

So,

$$\lim_{n \rightarrow \infty} [R_n(s, 0) - R_n(s + 1, 0)] = -\frac{1 - X}{X}(\beta^s - \beta^{s+1})K_0 > 0.$$

Next, it follows from 1. that $V_{T,n}^B(1, p_{0,n}) = V_{T,n}^B(\emptyset, p_{0,n})$ for all n . Thus by the same reasoning as above,

$$\lim_{n \rightarrow \infty} [R_n(T, 0) - R_n(\emptyset, 0)] = -\beta^T K_0 \left(\frac{1 - X}{X} \right) > 0.$$

3. **Show** $\lim_{n \rightarrow \infty} \mathbf{R}_n(t, 1) = 1$. For the equilibrium $(p_{t,n}^*, A_{t,n}^B)_{t=1}^T$ under $p_{0,n}$, let $H_{t,n}$ denote the good agent's distribution of time- t beliefs given $\tau \notin \{1, \dots, t\}$. Define

$$Q_{n,t} \equiv \frac{\int_0^1 \int_{p_{t,n}^*}^1 (1 - p_t) dG_t(p_t | p_{t-1}) dH_{t,n}(p_{t-1})}{\int_0^1 \int_{p_{t,n}^*}^1 p_t dG_t(p_t | p_{t-1}) dH_{t,n}(p_{t-1})}.$$

I begin by showing

$$\lim_{n \rightarrow \infty} \left(\frac{p_{0,n}}{1 - p_{0,n}} \right) Q_{n,t} = 0. \quad (27)$$

Since $\lim_{n \rightarrow \infty} p_{0,n} = 0$, it suffices to show that there exists $L \in \mathbb{R}^+$ such that $Q_{n,t} < L$ for all n . Note that for all n ,

$$\int_0^1 \int_{p_{t,n}^*}^1 p dG_t(p_t | p_{t-1}) > \int_0^1 \int_{p_{t,n}^*}^1 p_{t,n}^* dG_t(p_t | p_{t-1}).$$

Thus,

$$Q_{n,t} < \frac{\int_0^1 \int_{p_{t,n}^*}^1 dG_t(p_t | p_{t-1}) dH_{t,n}(p_{t-1})}{\int_0^1 \int_{p_{t,n}^*}^1 p_{t,n}^* dG_t(p_t | p_{t-1}) dH_{t,n}(p_{t-1})} - 1 = \frac{1}{p_{t,n}^*} - 1.$$

By the assumption that $p_{t,n}^* \geq \hat{p}_t$ for all n , $Q_{n,t} < \frac{1}{\hat{p}_t} - 1$, thus establishing (27). Now, recall that

$$R_n(t, \theta = 0) = \frac{1}{1 + \left(\frac{1-R_{t-1,n}}{R_{t-1,n}}\right) \left(\frac{A_{t,n}^B}{\int_0^1 \int_{p_{t,n}^*}^1 \frac{1-p_t}{1-p_{0,n}} dG_t(p_t | p_{t-1}) dH_{t-1,n}(p_{t-1})}\right)} \quad (28)$$

$$R_n(t, \theta = 1) = \frac{1}{1 + \left(\frac{1-R_{t-1,n}}{R_{t-1,n}}\right) \left(\frac{A_{t,n}^B}{\int_0^1 \int_{p_{t,n}^*}^1 \frac{p_t}{p_{0,n}} dG_t(p_t | p_{t-1}) dH_{t-1,n}(p_{t-1})}\right)}.$$

It follows from 2. that

$$\frac{A_{t,n}^B}{\int_0^1 \int_{p_{t,n}^*}^1 \frac{1-p_t}{1-p_{0,n}} dG_t(p_t | p_{t-1}) dH_{t-1,n}(p_{t-1})} \text{ converges.} \quad (29)$$

Now note

$$\begin{aligned} & \lim_{n \rightarrow \infty} \left(\frac{1-R_{t-1,n}}{R_{t-1,n}} \right) \left(\frac{A_{t,n}^B}{\int_0^1 \int_{p_{t,n}^*}^1 \frac{p_t}{p_{0,n}} dG_t(p_t | p_{t-1}) dH_{t-1,n}(p_{t-1})} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{A_{t,n}^B}{\int_0^1 \int_{p_{t,n}^*}^1 \frac{1-p_t}{1-p_{0,n}} dG_t(p_t | p_{t-1}) dH_{t-1,n}(p_{t-1})} \right) \left(\frac{\int_0^1 \int_{p_{t,n}^*}^1 \frac{1-p_t}{1-p_{0,n}} dG_t(p_t | p_{t-1}) dH_{t-1,n}(p_{t-1})}{\int_0^1 \int_{p_{t,n}^*}^1 \frac{p_t}{p_{0,n}} dG_t(p_t | p_{t-1}) dH_{t-1,n}(p_{t-1})} \right) = 0 \end{aligned}$$

where the final equality follows from (27) and (29). Thus, by (28),

$$\lim_{n \rightarrow \infty} R_n(t, 1) = 1.$$

4. **For some n , $\mathbf{V}_{t,n}^{\mathbf{R},\mathbf{G}}(1, \hat{\mathbf{p}}_t) > \mathbf{V}_{t,n}^{\mathbf{R},\mathbf{G}}(\emptyset, \hat{\mathbf{p}}_t)$.** For all n ,

$$V_{t,n}^{R,G}(1, \hat{p}_t) = \hat{p}_t R_n(t, \theta = 1) + (1 - \hat{p}_t) R_n(t, \theta = 0).$$

It thus follows from 2. and 3. above, and the fact that $\hat{p}_t < 1$, that there exists $K > 0$ and $N \in \mathbb{N}$ such that if $n > N$,

$$V_{t,n}^{R,G}(1, \hat{p}_t) > \hat{p}_t + (1 - \hat{p}_t)R_n(\tau, 0) + K \quad (30)$$

for $\tau \equiv \min\{t + 1, T\}$. Now, note that for any n ,

$$V_{t,n}^{R,G}(1, \hat{p}_t) \leq \hat{p}_t \max_{\tau \in \{t+1, \dots, T, \emptyset\}} R_n(\tau, \theta = 1) + (1 - \hat{p}_t) \max_{\tau \in \{t+1, \dots, T, \emptyset\}} R_n(\tau, \theta = 0).$$

Thus, by 2., there exists $N' \in \mathbb{N}$ such that for all $n \geq N'$

$$V_{t,n}^{R,G}(\emptyset, \hat{p}_t) \leq \hat{p}_t + (1 - \hat{p}_t)R_n(\tau, \theta = 0).$$

Thus, by (30), for all $n \geq \max\{N, N'\}$,

$$V_{t,n}^{R,G}(1, \hat{p}_t) > V_{t,n}^{R,G}(\emptyset, \hat{p}_t).$$

By Lemma 3 it follows from 4. that for all $n \geq \max\{N, N'\}$, $p_{t,n}^* < \hat{p}_t$, contradicting the assumption that $p_{t,n}^* \geq \hat{p}_t$ for all n .

Next, suppose $X = 1$. It follows directly from Lemma 7 that there exists a $\bar{p} \in (0, 1)$ such that if $p_0 < \bar{p}$, $p_t^* < \hat{p}_T \leq \hat{p}_t$ for all t under any equilibrium strategy $(p_t^*)_{t=1}^T$ under p_0 .

□

Proof of Theorem 2. Fix any equilibrium $(p_t^*, A_t^B)_{t=1}^T$ and any $t \in \{1, \dots, T\}$. First, consider the case where $A_s^B = 1$ for some $s < t$. Since G plays an interior cutoff strategy at all t , the equilibrium reputation function R must be such that

$$R(\tau, \theta) = 1 \text{ for all } \tau \in \{t, \dots, T, \emptyset\}, \theta \in \{0, 1\}.$$

Thus, $V_t^{R,G} = 1$ for all $a \in \{\emptyset, 1\}$, $p \in [0, 1]$. Hence, G 's problem at time t is to choose a strategy which maximizes the following:

$$E_\theta[U(\tau, \theta)] = (1 - X)\beta^\tau(K_\theta \mathbb{I}(\tau \neq \emptyset)) + X.$$

This problem is equivalent to maximizing $\beta^\tau(K_\theta \mathbb{I}(\tau \neq \emptyset))$. Hence, the equilibrium strategy must be equal to the optimal cutoff rule under $X = 0$, i.e., $p_t = \hat{p}_t$.

Next, consider the case where $A_s^B < 1$ for all $s < t$. I claim that in this case $p_t^* > \hat{p}_t$. First, suppose that $A_t^B = 1$. The equilibrium reputation function must be such that (1) $R(s, \theta) = 1$ for all $s \in \{t+1, \dots, T, \emptyset\}$ and $\theta \in \{0, 1\}$ and (2) $R(t, \theta) < 1$ for $\theta \in \{0, 1\}$. Together, these two facts imply that $V_t^{R,G}(1, p) < 1$ and $V_t^{R,G}(\emptyset, p) = 1$ for all p . Furthermore, by the same reasoning as above, $p_s^* = \hat{p}_s$ for all $s > t$ and thus $V_t^{NR,G}(a, p) = \hat{V}_t(a, p)$ for all a, p . Thus,

$$V_t^G(1, \hat{p}_t) = (1-X)V_t^{NR,G}(1, \hat{p}_t) + XV_t^{R,G}(1, \hat{p}_t) < (1-X)V_t^{NR,G}(\emptyset, \hat{p}_t) + XV_t^{R,G}(\emptyset, \hat{p}_t) = V_t^G(\emptyset, \hat{p}_t),$$

and thus $p_t^* > \hat{p}_t$. Next, suppose that $A_t^B < 1$. It must also be that $A_t^B > 0$. To show this, suppose not by contradiction. Then, the reputation function must be such that $R(t, \theta) = 1$ for $\theta \in \{0, 1\}$. Thus, $V_t^{R,B}(1, p) \geq V_t^{R,B}(\emptyset, p)$ for all p . Since $V_t^{NR,B}(1, p_0) > V_t^{NR,B}(\emptyset, p_0)$, it follows that $V_t^B(1, p_0) > V_t^B(\emptyset, p_0)$, and thus $A_t^B = 1$. Contradiction. So, $A_t^B \in (0, 1)$ which implies B must be indifferent at p_0 : $V_t^B(1, p_0) = V_t^B(\emptyset, p_0)$. Since $V_t^G(\emptyset, p_0) \geq V_t^B(\emptyset, p_0)$ and $V_t^B(1, p_0) = V_t^G(1, p_0) \leq V_t^G(\emptyset, p_0)$, $V_t^G(\emptyset, p_0) \geq V_t^G(1, p_0)$. It follows that $p_t^* \geq p_0 > \hat{p}_t$. $\square$

To prove [Theorem 3](#), I state and prove three lemmas.

Lemma 8. *Fix all parameters except β and X . There exists an $\underline{X}$ such that if $X > \underline{X}$, $A_t^B \in (0, 1)$ in any equilibrium, under any β .*

Proof. I begin by showing $A_t^B < 1$ for any t . To this end, I will prove the stronger statement that there exists $\underline{X} \in (0, 1)$ and $\underline{b} \in (0, 1)$ such that $A_t^B < \underline{b}$ in any equilibrium under any β when $X > \underline{X}$.

Proof by induction. Fix any t . Suppose there exists an $\underline{X}_{t-1}$ and $\underline{b}_{t-1} \in (0, 1)$ such that $A_s^B < \underline{b}_{t-1}$ for all $s < t$ when $X > \underline{X}_{t-1}$. Note this holds vacuously when $t = 1$. How, we want to show that there exists $\underline{X}_t \in (\underline{X}_{t-1}, 1)$ and $\underline{b}_t$ such that $A_s^B < \underline{b}_t$ for all $s \leq t$ when $X > \underline{X}_t$. Suppose not, by contradiction. Then, there exists a sequence of $X, (X_n)_{n=1}^\infty$ with $X_n \rightarrow 1$ where $A_t^n \rightarrow 1$ for some equilibrium strategy A_t^n for B under some value of β . Let superscript n denote equilibrium under each such equilibrium. I claim $V_t^{n,B}$ must satisfy the following two properties:

1. There exists an N and $\underline{V} < 1$ such that $V_t^{n,B}(p_0, 1) < \underline{V}$ for all $n > N$.

$$2. \lim_{n \rightarrow \infty} V_t^{n,B}(p_0, \emptyset) = 1.$$

Note that these two properties imply that there exists an N such that for all $n > N$, $V_t^{n,B}(p_0, 1) < V_t^{n,B}(p_0, \emptyset)$, thus implying that $A_t^{n,B} = 0$, a contradiction.

I begin by establishing 1. The inductive assumption implies there exists $\hat{R} < 1$ such that $R_{t-1}^n < \hat{R}$. Since $X_n \rightarrow 1$, it suffices to show that there exists N hat $\bar{R} < 1$ such that $R^n(t, 1) < \bar{R}$ for $n > N$. Since $A_t^n \rightarrow 1$, there exists an N and $\underline{A} > 0$ such that for all $n > N$, $A_t^n > \underline{A}$. Recall that

$$R^n(t, 1) = \frac{1}{1 + \frac{1-R_{t-1}}{R_{t-1}} \frac{p_0 A_t^{n,B}}{Pr^n(\tau=t|i=G, \theta=1)}}.$$

It thus follows that for all $n > N$,

$$R^n(t, 1) < \bar{R} \equiv \frac{1}{1 + \frac{1-\hat{R}}{\hat{R}} \underline{A} p_0} < 1.$$

Next, I establish 2. It suffices to show $R_t^n \rightarrow 1$. We know by assumption that $A_t^{B,n} \rightarrow 1$. Note that for any n such that $A_t^{B,n} = 1$, by equilibrium selection, $p_t^{*,n} < 1$, and thus $R_t^n = 1$. Let X_m denote the subsequence of $(X_n)_{n=1}^\infty$ such that $A_t^n < 1$. Note that if X_m is finite, then the statement holds trivially.

Now suppose X_m is an infinite subsequence. $A_t^m < 1$ implies $V_t^{B,m}(p_0, 1) \leq V_t^{B,m}(p_0, \emptyset)$ and thus, $V_t^{B,m} \leq V_t^{B,m}(p_0, \emptyset)$. This implies $V_t^{G,m} \leq V_t^{G,m}(p_0, \emptyset)$ and thus $p_t^{*,m} \geq p_0 > 0$. This further implies that there exists $q > 0$ such that $Pr^m(\tau \neq t | \tau \geq t, G) > q$. Furthermore, by the inductive assumption there exists a $\hat{R} > 0$ such that $R_{t-1}^m < \hat{R}$. Thus,

$$\lim_{m \rightarrow \infty} R_t^m = \lim_{m \rightarrow \infty} \frac{1}{1 + \frac{1-R_{t-1}^m}{R_{t-1}^m} \frac{1-A_t^m}{Pr^m(\tau \neq t | \tau \geq t, i=G)}} = 1.$$

Now, I show that $A_t^B > 0$ for any t . Suppose not, by contradiction. Then there exists an n and sequence of X $(X_n)_{n=1}^\infty$ with $X_n \rightarrow 1$ where $A_t^{n,B} = 0$ for all n under some equilibrium strategy $A_t^{B,n}$ under X_n and some β . Since G acts with positive probability in equilibrium under any t by [Lemma 2](#), $R^n(t, \theta) = 1$ for all θ . Since $X_n \rightarrow 1$, $\lim_{n \rightarrow \infty} V_t^{B,n}(p_0, 1) = 1$. Thus, B 's ex-ante reputational value $V_1^{B,R,n}(p_0) \rightarrow 1$. Since $R_0 < 1$, this implies that for n sufficiently large, B 's expected

reputation at the end of the game exceeds R_0 . Thus R^n cannot be consistent with Bayes Rule. Contradiction. $\square$

Lemma 9. Fix all parameters except β and X . Suppose $X > \underline{X}$, where $\underline{X}$ is established in Lemma 8. For any t , if there exists $\bar{p}_t$ such that $p_t^* < \bar{p}_t$ in all equilibrium for any β , there exists $\underline{R} > 0$ and $\bar{R} < 1$ such that $R(t, 0) \geq \underline{R}$ and $R(t, 1) \leq \bar{R}$.

Proof. I begin by establishing the bound $\underline{R}$. by Lemma 8, in every equilibrium under any β and t

$$V_t^B(p_0) = V_t^B(p_0, \emptyset).$$

Since $V_t^B(p_0, 1) = V_t^G(p_0, 1)$ and $V_t^B(p_0, \emptyset) \leq V_t^G(p_0, \emptyset)$, $V_t^G(p_0, 1) \leq V_t^G(p_0, \emptyset)$, and thus $p_t^* \geq p_0$ for all t . Thus there exists $\underline{p} \in (0, 1)$ such that $Pr(\tau = t | \theta = 0, i = G) \geq \underline{p}$ in any equilibrium, under any β . Namely, $\underline{p}$ is the probability $\tau = t$ given $\theta = 0$, $i = G$ and the good agent plays cutoff strategies $p_s = p_0$ for all $s < t$ and $p_t = \bar{p}_t$. Thus, in any equilibrium

$$R(t, 0) \geq \underline{R} \equiv \frac{1}{1 + \frac{1-R_0}{R_0} \frac{1}{\underline{p}}} > 0.$$

Now, I establish the bound $\bar{R}$. Suppose by contradiction that there does not exist such a $\bar{R}$. Then, there exists a sequence of equilibria, indexed by n (under possibly different values of β) such that

$$\lim_{n \rightarrow \infty} R^n(t, 1) = 1. \quad (31)$$

Now, recall that

$$R^n(t, 1) = \frac{1}{1 + \frac{1-R_0}{R_0} \frac{Pr^n(\tau=t|i=B, \theta=1)}{Pr^n(\tau=t|i=G, \theta=1)}}.$$

Thus, in order for (31), it must be that

$$\lim_{n \rightarrow \infty} Pr^n(\tau = t | i = B, \theta = 1) = 0.$$

Since $Pr^n(\tau = t | i = B, \theta = 0) = \frac{1-p_0}{p_0}$, $Pr^n(\tau = t | i = B, \theta = 1)$,

$$\lim_{n \rightarrow \infty} Pr^n(\tau = t | i = B, \theta = 0) = 0.$$

However, as previously shown for all n , there exists $\underline{p} > 0$ such that

$$Pr^n(\tau = t | \theta = 0, i = G) \geq \underline{p}.$$

This implies

$$\lim_{n \rightarrow \infty} \frac{Pr^n(\tau = t | \theta = 0, i = B)}{Pr^n(\tau = t | \theta = 0, i = G)} = 0,$$

and therefore $\lim_{n \rightarrow \infty} R^n(t, 0) = 1$. Since $\lim_{n \rightarrow \infty} R^n(t, \theta) = 1$ for all θ , it follows that for n sufficiently large, B can profitably deviate by playing the pure strategy $A_t^{B,n} = 1$. Contradiction. $\square$

Lemma 10. Fix all parameters except β and X . There exists $\underline{X} > 0$ such that if $X > \underline{X}$, then for all t , there exists $\bar{p}_t \in (0, 1)$ such that $p_t^* < \bar{p}_t$ in all equilibria.

Proof. Suppose $X > \underline{X}$, where $\underline{X}$ is the bound established in Lemma 8. I will now prove that for all t , there exists $\bar{p}_t \in (0, 1)$ such that $p_t^* < \bar{p}_t$ in all equilibria. Proof by (backwards) induction.

Base case: By Lemma 8, $p_t^* = p_0$. By Lemma 9, this implies that there exists $\underline{R}_T$ such that $R(T, 0) > \underline{R}_T$.

Induction step: Fix a $t \in \{1, \dots, T\}$. Assume for all $s > t$, there exists such a $\bar{p}_s \in (0, 1)$. By Lemma 9, for all $s > t$, there exists an $\underline{R}_s > 0$ such that $R(s, 0) > \underline{R}_s$ and $R(s, 1) < \bar{R}_s$. We want to show that there exists a $\bar{p}_t$ satisfying the above condition.

Suppose not, by contradiction. Then, there exists a sequence of $X \{X_n\}_{n=1}^\infty$ with $\lim_{n \rightarrow \infty} X_n = 1$ such that $\lim_{n \rightarrow \infty} p_t^{*,n} \rightarrow 1$, where $p_t^{*,n}$ is some equilibrium under $p_t^{*,n}$ for some β .

First, consider the case where $\lim_{n \rightarrow \infty} R^n(t, 1) = 1$. By Lemma 9, there exists a $\bar{R}_{t+1} < 1$ such that $R^n(t+1, 1) < \bar{R}_{t+1}$ for any n . By the continuity of the V^n , there exists $\bar{p}_t$ such that $p_t^* < \bar{p}_t$ for all n . Contradiction.

Next, consider the case where $R^n(t, 1)$ does not converge to 1. Then, there exists an infinite subsequence indexed by m and an upper bound $\bar{R}$ such that for all m , $R^m(t, 1) < \bar{R}$. Recall by the definition of R^m , there exists $\underline{p}$ such that for all m ,

$$\frac{Pr^m(\tau | i = B, \theta = 1)}{Pr^m(\tau | i = G, \theta = 1)} > \underline{p}.$$

Since $\lim_{m \rightarrow \infty} p_t^{*,m}$,

$$\lim_{m \rightarrow \infty} \frac{Pr^m(\tau|G, \theta = 1)}{Pr^m(\tau|G, \theta = 0)} = \infty.$$

Thus,

$$\lim_{m \rightarrow \infty} \frac{Pr^m(\tau|B, \theta = 0)}{Pr^m(\tau|G, \theta = 0)} = 1,$$

and thus $\lim_{m \rightarrow \infty} R^m(t, 0) = 0$. Note that by B 's indifference condition, for all $s > t$,

$$(1 - p_0)R^m(t, 0) + p_0R^m(t, 1) = (1 - p_0)R^m(s, 0) + pR^m(s, 1),$$

Since there exists a $\underline{R}$ such that $R^m(s, 0) > \underline{R}$ for all m , it follows that there exists an M and $\Delta > 0$ such that for all $m > M$,

$$R^m(t, 1) > \Delta + R^m(s, 1) \text{ for all } s > t.$$

Thus, it again follows that from the continuity of V^m that there exists $\bar{p}_t$ such that $p_t^{*,m} < \bar{p}_t$ for all $m > M$. Contradiction. $\square$

Proof of Theorem 3. To prove the first bullet, note that it follows from Lemma 10 that for all t , there exists a $\underline{X} \in (0, 1)$ and $\bar{p}_t \in (0, 1)$ such that $p_t^* < \bar{p}_t$ under any equilibrium for any β when $X > \underline{X}$. Fix this $\underline{X}$ and these $\bar{p}_t$. Now, note that for any $t < T$, fixing all other parameters (except X and β) there exists $\underline{\beta}_t \in (0, 1)$ such that if $\beta > \underline{\beta}_t$, $\hat{p}_t > \bar{p}_t$. Thus, for any $t < T$ and for all $X > \underline{X}$ and $\beta \geq \max_{s \leq t} \underline{\beta}_s$, $\hat{p}_t > p_t^*$.

To prove the second bullet, it suffices to show $p_T^* > p_T$. Note that there exists $\tilde{X} \in (0, 1)$ such that if $X > \tilde{X}$, $A_t^B \in (0, 1)$ for all t , in any equilibrium. Thus, by the same reasoning presented in the proof of Theorem 2, since $\hat{p}_t < p_0$, $p_T^* > p_T$. $\square$